

ZITLAS

AI MUSCLE GAIN BRAIN

VOLUME 1

RAG Knowledge Base for ZITLAS AI Fitness Coach
Personalized Muscle Gain · Diet Plans · Training Logic · Coaching Rules

ATTRIBUTE	VALUE
DOCUMENT TYPE	AI Training Knowledge Base
VERSION	1.0 2025
MODULES	13 Core Modules
COACHING RULES	300+ IF-THEN Rules
USE CASE	RAG · LLM Coach · Personalized Plans
CLASSIFICATION	ZITLAS Internal – Coach System

Atharva Sankpal (King) · CEO & Managing Director, ZITLAS
Amol Sangle · COO, ZITLAS

TABLE OF CONTENTS

MODULE 1

Hypertrophy, progressive overload, beginner gains, realistic expectations

MODULE 2

BMR, TDEE, surplus strategies, decision trees

MODULE 3

Amino acids, leucine threshold, protein quality, timing, targets

MODULE 4

Glycogen, carb timing, pre/post-workout, food sources

MODULE 5

Dietary fats, testosterone, EFAs, intake ranges

MODULE 6

Meal frequency, pre/post-workout, night nutrition, protein distribution

MODULE 7

100+ IF-THEN rules for athlete types, budgets, lifestyles

MODULE 8

Volume, frequency, intensity, compound/isolation, splits, key lifts

MODULE 9

Sleep, stress, deloads, CNS fatigue, overtraining

MODULE 10

Evidence-based only: creatine, whey, fish oil, caffeine

MODULE 11

Skinny beginners, hardgainers, women, vegetarians, hostel students

MODULE 12

Errors, correction strategies, troubleshooting

MODULE 13

300+ rules, decision logic, plateau handling, progress tracking

MODULE 1

MUSCLE GAIN FUNDAMENTALS

Core definitions, mechanisms, and expectations for the AI coach system

TERM	DEFINITION	COACHING RELEVANCE
Muscle Gain	Increase in skeletal muscle cross-sectional area and/or density through resistance training + nutrition	Primary goal variable
Hypertrophy	Enlargement of muscle fibers (myofibrils, sarcoplasm) due to mechanical tension, metabolic stress, or muscle damage	Triggered by resistance training
Myofibrillar Hypertrophy	Growth of contractile proteins (actin, myosin). Increases strength and muscle density.	Heavy loads (1–6 RM range)
Sarcoplasmic Hypertrophy	Increase in non-contractile fluid/glycogen within muscle cell. Increases muscle size more than strength.	Moderate loads (8–15 RM range)
Muscle Gain vs Weight Gain	Weight gain = any mass increase (fat, water, muscle, bone). Muscle gain = lean tissue increase only.	Track body composition, not just scale
Lean Bulk	Controlled calorie surplus (+200–400 kcal/day). Maximizes muscle gain, minimizes fat gain.	Recommended for most athletes
Dirty Bulk	Aggressive calorie surplus (+500–1000+ kcal/day). Faster weight gain but significant fat accumulation.	Use only for severely underweight
Body Recomposition	Simultaneous fat loss + muscle gain. Possible for beginners, detrained, obese, and enhanced athletes.	Requires slight deficit or maintenance + high protein
Progressive Overload	Systematically increasing training stimulus over time (load, volume, density, difficulty)	Core mechanism for continued adaptation
Beginner Gains	Rapid strength + muscle gains in first 6–12 months of training due to neural adaptation + low training history	Set high expectations for beginners
Training Adaptation	Body's physiological response to training stress. Requires adequate stimulus, nutrition, and recovery.	Adaptation = stress + recovery + nutrition

Three Primary Drivers

MECHANISM	DESCRIPTION	HOW TO TARGET	OPTIMAL REP RANGE
Mechanical Tension	Force generated in muscle under load. Primary driver of hypertrophy. Activates mTOR pathway.	Heavy compound lifts, slow eccentrics, full ROM	3–8 reps @ 70–85% 1RM
Metabolic Stress	Accumulation of metabolites (lactate, H+). Creates 'pump'. Secondary hypertrophy driver.	High rep sets, short rest, occlusion training	12–20 reps, short rest 60s
Muscle Damage	Micro-tears in muscle fibers. Triggers repair and growth via satellite cells.	Eccentric focus, novel movements, stretch-loaded exercises	8–15 reps with controlled eccentric

FIBER TYPE	CHARACTERISTICS	TRAINING RESPONSE	HYPERTROPHY POTENTIAL
Type I (Slow Twitch)	High endurance, low force, oxidative metabolism, fatigue resistant	Responds to high volume, low load	Low-moderate

FIBER TYPE	CHARACTERISTICS	TRAINING RESPONSE	HYPERTROPHY POTENTIAL
Type IIa (Fast Twitch)	Mixed properties, moderate force and endurance	Responds to moderate-heavy loads, moderate volume	High
Type IIx (Fast Twitch)	High force, low endurance, glycolytic, fatigue quickly	Responds to heavy loads, low reps, explosive training	Highest

Most muscles are mixed fiber types. Training multiple rep ranges optimizes hypertrophy across fiber types.

STRATEGY	CALORIE SURPLUS	RATE OF GAIN	FAT GAIN RISK	BEST FOR
Lean Bulk	+200–350 kcal/day	0.25–0.5 kg/month	Low	Most intermediate athletes, BF 10–18% (M) / 18–28% (F)
Moderate Bulk	+300–500 kcal/day	0.5–1.0 kg/month	Moderate	Beginners, underweight athletes
Aggressive Bulk	+500–1000 kcal/day	1.0–2.0+ kg/month	High	Severely underweight (BMI <18.5), hardgainers
Dirty Bulk	+1000+ kcal/day	2+ kg/month	Very High	Not recommended. Short-term only if severely underweight
Body Recomp	±0 (maintenance)	Muscle +, Fat –	None	Beginners, returning athletes, BF >20% (M) / >28% (F)
Mini Cut-Bulk Cycle	Alternating phases	Controlled	Minimal	Advanced athletes managing body fat

Methods of Progressive Overload

METHOD	DESCRIPTION	EXAMPLE	PRIORITY
Load Increase	Add weight to the bar/machine	+2.5 kg when hitting top of rep range	PRIMARY
Volume Increase	Add sets or reps	4x8 → 4x10 → 5x10	PRIMARY
Density Increase	Same volume in less time	Rest 90s → 60s	SECONDARY
Range of Motion	Increase ROM on movements	Parallel squat → full depth squat	SECONDARY
Tempo Manipulation	Slower eccentrics increase time under tension	2-0-1 → 3-1-1 tempo	SECONDARY
Exercise Difficulty	Progress to harder variation	Push-up → Incline Press → Bench Press	TERTIARY
Frequency	Train muscle group more often	1x/week → 2x/week	TERTIARY

PHASE	DURATION	PRIMARY ADAPTATION	EXPECTED MUSCLE GAIN
Beginner	0–12 months	Neural adaptation (strength↑↑), initial hypertrophy	6–12 kg muscle/year
Intermediate	1–3 years	Hypertrophy dominant, strength plateaus	2–4 kg muscle/year
Advanced	3–5+ years	Slow hypertrophy, refinement	0.5–1.5 kg muscle/year
Elite	5+ years	Marginal gains	<0.5 kg muscle/year

ATHLETE PROFILE	MONTHLY GAIN (kg)	ANNUAL GAIN (kg)	NOTES
Beginner Male (natural)	0.8–1.5	9–18	Neural gains + muscle simultaneously
Beginner Female (natural)	0.4–0.8	4–9	Proportionally similar % but lower absolute

Athlete Profile	Monthly Gain (kg)	Annual Gain (kg)	Notes
Intermediate Male	0.2–0.5	2–5	Requires consistent progressive overload
Intermediate Female	0.1–0.3	1–3	Focus on strength benchmarks
Advanced Male	0.05–0.15	0.5–1.5	Diminishing returns phase
Advanced Female	0.03–0.1	0.3–1	Requires precise program + nutrition

Expectation-setting is critical to prevent program hopping and dropout. Brief athlete on realistic timelines at onboarding.

- [R001] Set muscle gain expectations during onboarding. Show realistic monthly and annual gain targets.
- [R002] Classify athlete as beginner/intermediate/advanced before generating any training or nutrition plan.
- [R003] If athlete has <1 year training history, classify as beginner. Apply beginner protocols.
- [R004] Never recommend dirty bulk unless athlete is severely underweight (BMI <17) AND has high caloric expenditure.
- [R005] Recommend lean bulk (+200–350 kcal) as default for athletes with BF 10–20% (M) or 18–28% (F).
- [R006] If athlete BF >20% (M) or >28% (F), do NOT recommend calorie surplus. Use recomp or mild deficit.
- [R007] Progressive overload must be the core principle of every training plan generated.
- [R008] If athlete is beginner, prioritize load progression over volume manipulation.
- [R009] Body recomposition is a valid strategy for beginners, returning athletes, and obese individuals.
- [R010] Training adaptation requires: adequate stimulus + adequate nutrition + adequate recovery. All three must be present.

MODULE 2

CALORIE SURPLUS SYSTEM

BMR, TDEE, maintenance, surplus strategies, and decision trees

Primary Formulas (AI Coach Uses Mifflin-St Jeor by Default)

FORMULA	MALE	FEMALE	ACCURACY
Mifflin-St Jeor (PREFERRED)	$(10 \times W) + (6.25 \times H) - (5 \times A) + 5$	$(10 \times W) + (6.25 \times H) - (5 \times A) - 161$	Most accurate for general pop. ±5%
Harris-Benedict (Revised)	$(13.397 \times W) + (4.799 \times H) - (5.677 \times A) + 88.362$	$(9.247 \times W) + (3.098 \times H) - (4.330 \times A) + 447.593$	Slightly overestimates
Katch-McArdle	$370 + (21.6 \times \text{LBM})$	Same formula	Best for known body composition data
Simple Estimate	$W(\text{kg}) \times 22\text{--}24$ (sedentary)	$W(\text{kg}) \times 21\text{--}23$ (sedentary)	Quick estimation only

W = Weight (kg), H = Height (cm), A = Age (years), LBM = Lean Body Mass (kg)

TDEE = BMR × Activity Multiplier

ACTIVITY LEVEL	MULTIPLIER	DESCRIPTION	EXAMPLE
Sedentary	1.2	No exercise, desk job	Office worker, no gym
Lightly Active	1.375	1–3 days light exercise/week	Walks 30 min, gym 2x/week
Moderately Active	1.55	3–5 days moderate exercise/week	Gym 4x/week, active job
Very Active	1.725	6–7 days hard exercise/week	Daily training, physical job
Extra Active	1.9	2× daily training or very physical job	Athletes, construction workers
Adjustment: Add NEAT	+100–300 kcal	Non-exercise activity thermogenesis	Fidgeting, walking, housework

For Indian athletes: TDEE often overestimated. Use multiplier on lower end. Adjust based on 2-week weight trend.

GOAL	FORMULA	DAILY CALORIE TARGET	WEEKLY WEIGHT CHANGE
Maintenance	$TDEE \times 1.0$	TDEE	0 kg
Lean Bulk	$TDEE + 200\text{--}350$	$TDEE + 250$ (avg)	+0.1–0.25 kg/week
Moderate Bulk	$TDEE + 350\text{--}500$	$TDEE + 400$ (avg)	+0.25–0.5 kg/week
Aggressive Bulk	$TDEE + 500\text{--}1000$	$TDEE + 700$ (avg)	+0.5–1.0 kg/week
Cut (while bulking)	$TDEE - 300\text{--}500$	$TDEE - 400$ (avg)	–0.3–0.5 kg/week
Recomp	$TDEE \pm 0$	TDEE	~0 kg (composition shifts)

PROFILE	BMR	TDEE	LEAN BULK TARGET	MODERATE BULK TARGET
Male, 70kg, 175cm, 22yr, Mod Active	1,695 kcal	2,627 kcal	2,877 kcal	3,027 kcal

PROFILE	BMR	TDEE	LEAN BULK TARGET	MODERATE BULK TARGET
Male, 60kg, 170cm, 20yr, Lightly Active (Hostel)	1,544 kcal	2,123 kcal	2,373 kcal	2,523 kcal
Female, 55kg, 162cm, 21yr, Mod Active	1,356 kcal	2,102 kcal	2,302 kcal	2,452 kcal
Male, 85kg, 180cm, 28yr, Very Active	1,985 kcal	3,423 kcal	3,673 kcal	3,823 kcal
Female, 65kg, 168cm, 25yr, Lightly Active	1,468 kcal	2,019 kcal	2,219 kcal	2,369 kcal

Primary Decision: Which Surplus Strategy?

IF athlete is BEGINNER AND underweight (BMI <18.5) → THEN Use MODERATE BULK: TDEE +350–500 kcal

IF athlete is BEGINNER AND normal weight (BMI 18.5–24.9) → THEN Use LEAN BULK: TDEE +200–350 kcal

IF athlete is BEGINNER AND overweight (BMI >25) → THEN Use RECOMP: TDEE ±0, high protein 2g/kg

IF athlete is INTERMEDIATE AND BF <15% (M) / <23% (F) → THEN Use LEAN BULK: TDEE +200–300 kcal

IF athlete is INTERMEDIATE AND BF 15–20% (M) / 23–28% (F) → THEN Use LEAN BULK conservative: TDEE +150–200 kcal

IF athlete is INTERMEDIATE AND BF >20% (M) / >28% (F) → THEN STOP bulk. Run mini-cut first. TDEE –300 kcal

IF athlete is ADVANCED → THEN Use LEAN BULK only: TDEE +150–250 kcal. Slow gains expected.

IF athlete is HARDGAINER (gaining <0.1 kg/week on lean bulk) → THEN Increase by +150 kcal every 2 weeks until gaining

IF weight gain >0.6 kg/week on lean bulk → THEN Reduce calories by 150–200 kcal

IF weight stable for 2+ weeks on surplus → THEN Increase calories by 150–200 kcal

IF athlete has HIGH stress job / poor sleep → THEN Reduce surplus 100 kcal; focus on recovery first

IF athlete is hostel student with limited food access → THEN Calculate min calorie floor: BW(kg) × 35 kcal/kg

IF athlete is female AND reports menstrual irregularity on deficit → THEN STOP any deficit. Return to maintenance immediately.

IF athlete is vegetarian AND struggling to hit calories → THEN Add 30–50g peanut butter, 500ml whole milk to daily intake

MACRO	CALORIES/GRAM	MUSCLE GAIN %	KCAL EXAMPLE (3000 kcal)	GRAMS EXAMPLE
Protein	4 kcal/g	25–30%	750–900 kcal	188–225g
Carbohydrates	4 kcal/g	45–55%	1350–1650 kcal	338–413g
Fats	9 kcal/g	20–30%	600–900 kcal	67–100g

ATHLETE TYPE	TYPICAL TDEE	LEAN BULK TARGET	FOOD BUDGET ESTIMATE
Hostel Male, 60–70kg, Mod Active	2,000–2,500	2,250–2,750	■150–250/day
College Male, 65–75kg, Mod Active	2,200–2,700	2,450–2,950	■200–350/day
College Female, 50–60kg, Mod Active	1,700–2,100	1,900–2,350	■150–250/day
Working Professional Male, 70–85kg	2,200–2,800	2,450–3,050	■300–500/day
Working Professional Female, 55–70kg	1,800–2,200	2,000–2,450	■250–400/day
Athlete (high activity), Male	2,800–3,500	3,050–3,750	■400–700/day

- [R011] Always calculate TDEE before generating a calorie target. Never use generic calorie numbers.
- [R012] Default surplus recommendation: +250 kcal/day for lean bulk.
- [R013] Adjust calorie targets every 2 weeks based on actual weight trend, not estimates.
- [R014] If athlete gains >0.6 kg/week, reduce calories by 200 kcal to minimize fat gain.

- [R015] If athlete not gaining weight for 2 weeks, increase calories by 150–200 kcal.
- [R016] Minimum protein must be met FIRST before distributing remaining calories to carbs/fats.
- [R017] For hostel students: calculate minimum affordable calorie floor ($BW \times 33\text{--}35 \text{ kcal/kg}$).
- [R018] Never prescribe same calorie target for different body weights. Always personalize.
- [R019] Account for NEAT: active students/workers need higher TDEE multiplier.
- [R020] Hardgainers: increase calories by 150 kcal increments every 2 weeks until gaining 0.25 kg/week.

MODULE 3

PROTEIN SCIENCE

Amino acids, leucine threshold, protein quality, timing, and daily targets

Why Protein is the Critical Macronutrient for Muscle Gain

- Provides amino acids (building blocks) for muscle protein synthesis (MPS)
- Stimulates anabolic signaling pathways (mTORC1) via leucine
- Provides nitrogen for protein turnover and repair
- Has highest thermic effect of food (25–30% of calories burned in digestion)
- Preserves muscle mass during caloric deficits
- Most satiating macronutrient – important for body composition management

CATEGORY	AMINO ACIDS	DIETARY REQUIREMENT	COACHING NOTE
Essential Amino Acids (EAA)	Leucine, Isoleucine, Valine, Lysine, Methionine, Phenylalanine, Threonine, Tryptophan, Histidine	MUST come from diet	Deficiency stops MPS regardless of calorie surplus
Branched-Chain Amino Acids (BCAA)	Leucine, Isoleucine, Valine (subset of EAA)	Must come from diet	Leucine is the KEY anabolic trigger
Non-Essential Amino Acids (NEAA)	Alanine, Asparagine, Aspartic acid, Glutamic acid, Serine	Body can synthesize	Less critical for muscle gain
Conditionally Essential	Arginine, Cysteine, Glutamine, Glycine, Proline, Tyrosine	Required under stress/injury	Glutamine important for immune function during heavy training

The Leucine Trigger Rule

- Leucine is the PRIMARY anabolic amino acid. Acts as a sensor for mTORC1 pathway.
- Minimum leucine dose to maximally stimulate MPS: 2–3g leucine per meal
- Optimal leucine per meal: 3–4g for full MPS activation
- A 30–40g protein dose from high-quality protein delivers ~2.5–3.5g leucine
- Plant proteins: lower leucine density, require higher doses for same MPS response
- MPS stimulation lasts ~3–5 hours after a leucine-rich meal
- MPS is refractory: additional protein within 90 min of prior dose adds minimal benefit

PROTEIN SOURCE	LEUCINE % OF PROTEIN	PROTEIN NEEDED FOR 3g LEU	QUALITY NOTES
Whey Protein	10–11%	~27g	Best leucine source. Fast-absorbing.
Eggs (whole)	8.5%	~35g (6 eggs)	Complete EAA profile
Chicken Breast	8%	~38g	Lean, complete, high quality
Beef	8%	~38g	High in creatine + zinc
Fish (Tuna/Salmon)	8–9%	~33–38g	Salmon has omega-3 bonus
Paneer (cottage cheese)	7%	~43g	Key vegetarian source

PROTEIN SOURCE	LEUCINE % OF PROTEIN	PROTEIN NEEDED FOR 3g LEU	QUALITY NOTES
Milk (full fat)	~10% of protein	30g protein = 300ml whole milk per serve	Casein+whey blend
Soy Protein Isolate	7.5%	~40g	Best plant protein for MPS
Lentils / Dal	6–7%	~43–50g	Incomplete – combine with rice
Peanuts / PB	4%	~75g	Low leucine; supplement with dairy
Rice Protein	5%	~60g	Combine with pea protein
Pea Protein	6–8%	~38–50g	Good plant alternative

SCORING METHOD	DESCRIPTION	TOP SCORES
PDCAAS (Digestibility-Corrected Amino Acid Score)	Measures essential amino acid profile + digestibility. Max = 1.0	Whey: 1.0, Egg: 1.0, Milk: 1.0, Soy: 1.0, Beef: 0.92
DIAAS (Digestible Indispensable AA Score)	More accurate than PDCAAS. No correction ceiling.	Whey concentrate: 1.09, Whole egg: 1.13, Milk: 1.18, Soy: 0.91
BV (Biological Value)	Nitrogen retained/absorbed × 100	Whey: 104, Egg: 100, Casein: 77, Soy: 74, Wheat: 54

DIAAS is the modern gold standard. PDCAAS is still widely used. For RAG purposes: rank protein sources by DIAAS score.

FACTOR	ANIMAL PROTEIN	PLANT PROTEIN	COACHING IMPLICATION
Amino acid completeness	Complete (all EAAs in optimal ratio)	Usually incomplete (except soy, quinoa)	Vegetarians must combine proteins
Leucine content	High (8–11% of protein)	Lower (4–8%)	Plant protein dose must be 20–30% higher
Digestibility	85–99%	60–90%	Plant protein bioavailability lower
Nutrient co-factors	Vitamin B12, heme iron, zinc, creatine	Fibre, polyphenols, phytates	Vegetarians often deficient in B12, zinc
MPS per gram	Higher (more leucine)	Lower (less leucine)	Vegetarians need higher total protein target
Anti-nutritional factors	None significant	Phytates, lectins, trypsin inhibitors reduce absorption	Cooking and soaking reduce anti-nutrients
Cost (India)	■400–700/kg protein (chicken)	■100–300/kg protein (dal, soy)	Plant protein more economical

TIMING WINDOW	RECOMMENDATION	RATIONALE	EXAMPLE
Pre-Workout (60–90 min)	25–40g protein	Provides amino acids during training, reduces muscle breakdown	100g chicken + rice OR 1 scoop whey
Post-Workout (0–2 hrs)	30–50g fast protein	Elevates MPS during anabolic window	1–1.5 scoops whey OR 4 eggs + milk
Between meals	Every 3–5 hours	Maintain MPS stimulation throughout day	Protein distributed in 4–5 meals
Pre-Sleep	30–40g slow protein (casein)	Sustains overnight MPS, reduces muscle breakdown	200g paneer OR 1 cup Greek yogurt OR casein shake
Morning (within 60 min)	30–40g protein	Break overnight fast, initiate MPS	3–4 eggs + milk OR protein shake
Intra-Workout	EAAs (10–15g) if training >90 min	Prevents muscle breakdown in long sessions	EAA supplement or BCAA 5–8g

ATHLETE TYPE	MINIMUM (g/kg BW)	OPTIMAL (g/kg BW)	MAXIMUM USEFUL (g/kg BW)	NOTES
Beginner (any)	1.4	1.6–1.8	2.0	Newer trainees may benefit with lower intake
Intermediate	1.6	1.8–2.0	2.2	Higher end during caloric deficit
Advanced	1.8	2.0–2.2	2.4	High volume training demands more
Lean Bulk	1.6	1.8–2.0	2.2	Prioritize MPS with surplus
Cutting (deficit)	2.0	2.2–2.6	3.0	High protein preserves muscle in deficit
Recomp	1.8	2.0–2.4	2.8	Critical for simultaneous fat loss + muscle gain
Vegetarian	+10%	2.0–2.4 g/kg	2.8	Lower bioavailability requires more protein
Female athlete	1.4	1.6–2.0	2.2	Similar to male relative to LBM
Older adult (40+)	1.8	2.0–2.4	2.8	Anabolic resistance increases with age

FOOD	PORTION	PROTEIN (g)	COST (■)	TYPE
Eggs (whole)	6 eggs	36	30–40	Animal
Chicken breast	200g cooked	60	60–90	Animal
Tuna (canned)	150g	33	50–70	Animal
Paneer	200g	28	50–100	Vegetarian
Soy chunks (nutri-nuggets)	100g dry	52	15–25	Vegetarian
Whole milk	500ml	17	25–35	Vegetarian
Curd/Dahi	300g full fat	11	20–30	Vegetarian
Roasted chana	100g	17	15–25	Vegetarian
Moong dal	100g dry	25	20–35	Vegetarian
Rajma	100g dry	21	20–30	Vegetarian
Peanuts	100g	25	15–25	Vegetarian
Tofu	200g	17	40–70	Vegetarian
Whey protein (1 scoop)	30g powder	24–26	40–70	Supplement
Greek yogurt	200g	20	60–100	Vegetarian
Fish (Rohu/Pomfret)	200g	42	60–100	Animal

[R021] Default protein target for muscle gain: 1.8–2.0 g/kg bodyweight.

[R022] For vegetarians: increase protein target by 10–20% to account for lower bioavailability.

[R023] Each meal should contain 30–50g protein to maximally stimulate MPS.

[R024] Always include leucine-rich protein source (dairy, eggs, meat, whey) in post-workout meal.

[R025] Distribute protein evenly across 4–5 meals. Avoid front-loading or back-loading all protein.

[R026] Pre-sleep protein (casein/paneer) improves overnight MPS. Recommend to all bulking athletes.

[R027] If athlete cannot eat 4 protein meals, recommend protein supplement to bridge gap.

[R028] Plant protein sources must be combined to achieve complete amino acid profile (rice+lentils, corn+beans).

[R029] Protein above 3.0 g/kg shows no additional benefit for muscle gain. Flag as unnecessary cost.

[R030] For hostel students: eggs + milk + soy chunks = most cost-effective protein strategy.

MODULE 4

CARBOHYDRATES FOR MUSCLE GROWTH

Glycogen, carb timing, food sources, and performance optimization

- Primary fuel source for resistance training (glycolytic pathway)
- Replenish muscle and liver glycogen post-training
- Insulin spike post-exercise drives nutrient uptake into muscle cells
- Spare protein from oxidation (protein-sparing effect) — allows protein to be used for MPS not energy
- Support training intensity and volume — insufficient carbs = reduced performance
- Growth hormone release at night improved by normal carb intake
- Anti-catabolic effect: insulin is highly anabolic/anti-catabolic hormone

PARAMETER	MUSCLE GLYCOGEN	LIVER GLYCOGEN
Capacity	300–600g (1200–2400 kcal)	70–100g (280–400 kcal)
Function	Fuel for working muscle	Blood glucose maintenance
Depletion rate (high intensity)	~1–2g per min during intense training	~0.5–1g/hr during exercise
Time to full depletion	45–90 min of intense training	12–16 hrs fasting
Replenishment rate	~5–8g carbs/hr per kg body weight	Very fast with carb intake
Importance for hypertrophy	Critical: training intensity directly correlated with glycogen	Secondary

WINDOW	AMOUNT	TYPE	PURPOSE	EXAMPLES (INDIAN)
Pre-Workout (2–3 hrs)	1.5–2g/kg BW	Complex, low GI	Load glycogen, slow-release energy	Brown rice + dal, oats, banana + milk
Pre-Workout (30–60 min)	0.5–1g/kg BW	Simple/fast	Quick energy boost	Banana, dates, white bread + honey
Post-Workout (0–2 hrs)	1–1.5g/kg BW	High GI, fast	Rapidly replenish glycogen, spike insulin	White rice, bread, banana, fruit juice
Rest of the day	2–4g/kg BW (balance)	Mixed	General energy, glycogen maintenance	Chapati, rice, oats, sweet potato
Pre-Sleep	Moderate 50–100g	Moderate GI	Support overnight recovery, GH	Rice + dal, banana + milk
Intra-Workout (>90 min)	30–60g/hr	Simple + electrolytes	Sustain performance	Sports drink, dates, banana pieces

GOAL	CARB TARGET (g/kg BW)	DAILY EXAMPLE (70kg)	NOTES
Lean Bulk (moderate activity)	4–5 g/kg	280–350g	Fill remaining calories after protein + fats
Lean Bulk (high activity)	5–7 g/kg	350–490g	High volume athletes need more
Cutting (preserve muscle)	2–3 g/kg	140–210g	Maintain performance, reduce fat gain

GOAL	CARB TARGET (g/kg BW)	DAILY EXAMPLE (70kg)	NOTES
Recomp	3–4 g/kg	210–280g	Cycle carbs on training/rest days
Rest Day	2–3 g/kg	140–210g	Lower carbs okay on rest days
Training Day	5–7 g/kg	350–490g	Load up for performance

High Carb Sources – Ranked by Performance Value

FOOD	PORTION	CARBS (g)	GI	COST (■)	BEST USE
White Rice	200g cooked	62	High (72)	8–15	Post-workout, main meals
Brown Rice	200g cooked	46	Medium (50)	12–20	Pre-workout, general
Oats (rolled)	100g dry	66	Medium (55)	8–15	Breakfast, pre-workout
Banana	2 medium	54	Medium (51)	10–20	Pre/post-workout snack
Sweet Potato	200g	40	Medium (50)	15–25	Pre-workout, general
Chapati/Roti (wheat)	3 medium	60	Medium (62)	5–10	General meals
Bread (white)	4 slices	56	High (71)	10–20	Post-workout, quick carbs
Dates	100g (10 dates)	75	High (42–62)	30–60	Pre/post-workout boost
Moong Dal (cooked)	200g	28	Low (25)	15–25	General meals
Corn/Maize	200g	40	Medium (52)	10–20	General meals
Poha (flattened rice)	100g dry	77	Medium	8–15	Breakfast, pre-workout
Idli (2 pieces)	100g	22	Low-Medium	5–15	Breakfast

- [R031] Default carb recommendation for bulking athletes: 4–6 g/kg BW/day.
- [R032] NEVER restrict carbs to below 2 g/kg BW during muscle gain phase – performance will suffer.
- [R033] Post-workout meal MUST contain fast carbs (high GI) within 2 hrs for optimal glycogen resynthesis.
- [R034] Pre-workout meal (2 hrs prior): complex carbs + moderate protein. Avoid high fat.
- [R035] Carb cycling: higher carbs on training days, lower on rest days, keeping weekly average constant.
- [R036] For hostel/budget athletes: white rice + dal is the backbone cheap carb+protein combo.
- [R037] If athlete reports fatigue/low energy in training, increase carbs by 0.5 g/kg before reducing volume.
- [R038] Fiber target: 25–35g/day. Excessive fiber pre-workout causes GI distress – avoid 2 hrs pre-training.
- [R039] Intra-workout carbs only needed for sessions >90 minutes.
- [R040] South Indian athletes: rice-based diet is highly compatible with muscle gain when portioned correctly.

MODULE 5

HEALTHY FATS & HORMONAL SUPPORT

Dietary fat types, testosterone, recovery, brain function, essential fatty acids

- Testosterone and anabolic hormone synthesis requires dietary cholesterol and saturated/monounsaturated fats
- Cell membrane integrity and fluidity – critical for muscle cell signaling
- Fat-soluble vitamin absorption: Vitamins A, D, E, K require dietary fat
- Vitamin D (from sun + diet) is critical for testosterone production and muscle protein synthesis
- Omega-3 fatty acids (EPA/DHA) reduce inflammation, enhance MPS, improve insulin sensitivity
- Anti-inflammatory effects support faster recovery from resistance training
- Brain function and neurotransmitter production support training motivation
- Energy reserve: at 9 kcal/g, fats are calorie-dense – important for hard gainers

FAT TYPE	SOURCES	MUSCLE GAIN ROLE	RECOMMENDATION
Saturated Fat	Ghee, butter, coconut oil, red meat, eggs, whole milk	Testosterone precursor, cell membrane structure	15–20% of total fat intake. Include, don't eliminate.
Monounsaturated Fat (MUFA)	Olive oil, avocado, peanuts, almonds	Anti-inflammatory, heart health, hormone support	Primary fat source. 40–50% of fat intake.
Polyunsaturated Fat (PUFA) Omega-6	Sunflower oil, safflower oil, corn oil, seeds	Cell membrane, inflammatory signaling	Limit to 5–10% of fat intake. Most Indians overconsumed.
Polyunsaturated Fat (PUFA) Omega-3	Fatty fish (salmon, sardines), flaxseed, walnuts, chia, fish oil capsules	Anti-inflammatory, MPS enhancement, insulin sensitivity	PRIORITIZE. 2–4g EPA+DHA per day from food+supplements.
Trans Fats (Artificial)	Vanaspati, margarine, ultra-processed foods	Harmful. Reduces testosterone, promotes inflammation.	AVOID entirely. Zero tolerance.

Dietary Factors that Impact Testosterone

FACTOR	EFFECT	PRACTICAL RECOMMENDATION
Dietary fat intake	Low fat (<20% calories) reduces testosterone by 10–15%	Maintain fat at 20–35% of total calories
Saturated fat	Moderate intake supports Leydig cell testosterone production	2–3 tbsp ghee/week, eggs daily
Zinc	Zinc deficiency = testosterone deficiency. Critical cofactor.	Meat, seafood, pumpkin seeds, chickpeas. 11mg/day men, 8mg/day women
Vitamin D	Acts as steroid hormone. D deficiency = reduced testosterone.	Sun exposure 20–30 min/day. Supplement 1000–2000 IU/day if deficient.
Magnesium	Increases free testosterone. Often deficient in athletes.	Dark leafy greens, nuts, legumes. 400mg/day for athletes.
Omega-3	Reduces cortisol, may modestly increase testosterone	2–4g EPA+DHA/day from fish/fish oil
Alcohol	Acutely and chronically reduces testosterone	Advise complete avoidance during muscle gain phases

FACTOR	EFFECT	PRACTICAL RECOMMENDATION
Caloric deficit	Significant deficit (>500 kcal/day) reduces testosterone	Avoid prolonged deep deficits during muscle gain
Sleep deprivation	1 week of 5hr sleep reduces testosterone by 10–15%	Minimum 7–8 hours per night, mandatory

EFA	TYPE	DAILY TARGET	FOOD SOURCES	SUPPLEMENT
Omega-3 ALA	Plant-based precursor (inefficiently converted to EPA/DHA)	2–4g/day	Flaxseed, chia seeds, walnuts	Not prioritized
Omega-3 EPA	Anti-inflammatory, recovery	1–2g/day	Fatty fish (sardines, salmon, mackerel)	Fish oil 1–2g EPA
Omega-3 DHA	Brain function, cell membrane, recovery	0.5–1g/day	Fatty fish, algae oil	Fish oil or algae oil
Omega-6 LA (Linoleic Acid)	Cell function, inflammatory signaling	5–10g/day (already high in Indian diet)	Sunflower oil, corn oil	No supplement needed
Omega-9 MUFA (not essential)	Heart health, testosterone	Majority of fat intake	Olive oil, avocado, peanuts	No supplement needed

ATHLETE TYPE	FAT % OF CALORIES	GRAMS (2500 kcal)	GRAMS (3000 kcal)	NOTES
Lean Bulk (standard)	20–30%	56–83g	67–100g	Keep fats moderate, prioritize protein + carbs
High Fat Preference	30–35%	83–97g	100–117g	Lower carbs accordingly
Cutting phase	20–25%	56–69g	67–83g	Maintain fats for hormone support even in deficit
Hardgainer (low appetite)	30–35%	83–97g	100–117g	Calorie-dense fats help hit caloric targets
Minimum floor (hormonal)	>20%	>56g	>67g	NEVER go below 20% fat. Testosterone risk.

FOOD	PORTION	FAT (g)	TYPE	COST (■)
Whole eggs	3 eggs	15	MUFA + Sat + cholesterol	20–30
Ghee (desi)	1 tbsp (15g)	15	Saturated + MUFA	10–20
Groundnut oil	1 tbsp (15g)	15	MUFA + Omega-6	5–10
Mustard oil	1 tbsp (15g)	15	MUFA + Omega-3 ALA	5–10
Coconut oil	1 tbsp (15g)	15	Saturated (MCT)	10–20
Peanut butter	30g (2 tbsp)	16	MUFA + Omega-6	20–40
Almonds	30g (20 almonds)	14	MUFA + Vit E	30–60
Flaxseeds	30g (2 tbsp)	12	Omega-3 ALA	5–15
Walnuts	30g (7 halves)	18	Omega-3 ALA	30–60
Full fat milk	250ml	8	Saturated + MUFA	15–25
Fish oil capsule	1 capsule (1g)	1	EPA + DHA Omega-3	2–5
Sardines (canned)	100g	10	Omega-3 EPA+DHA	30–60

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- [R041] Minimum fat intake: 20% of total calories. Never prescribe below this – testosterone risk.
- [R042] Target 2–4g EPA+DHA omega-3 per day for all athletes. Supplement with fish oil if fish intake is low.
- [R043] Indian athletes: replace excess omega-6 vegetable oils (sunflower, corn) with mustard, groundnut, or olive oil.
- [R044] Hardgainers struggling to eat enough: increase fat intake to 30–35% of calories for caloric density.
- [R045] Zinc + Magnesium + Vitamin D: assess deficiency risk for all athletes. Supplement if dietary intake is low.
- [R046] Vegetarian athletes: use flaxseeds, walnuts, chia seeds for ALA omega-3. Consider algae oil for EPA+DHA.
- [R047] Ghee is culturally acceptable and nutritionally valid fat source. Recommend 1–2 tbsp/day within calorie budget.
- [R048] Avoid trans fats (vanaspati, dalda, processed biscuits) absolutely during training phases.
- [R049] Adequate dietary fat is essential for fat-soluble vitamin absorption (A, D, E, K).
- [R050] Pre-workout meal: minimize fat to <10g – fat slows gastric emptying and blunts energy delivery.

MODULE 6

MEAL TIMING SYSTEM

Meal frequency, pre/post workout, nighttime nutrition, protein distribution

MEAL FREQUENCY	MPS STIMULATION	PRACTICAL EASE	RECOMMENDATION
2 meals/day	Suboptimal – long periods of no MPS	Easy for some	NOT recommended for muscle gain. Protein under-distributed.
3 meals/day	Adequate if each meal has 40–50g protein	Most culturally normal	Minimum for muscle gain. Add 1–2 snacks.
4 meals/day	Good – consistent MPS every 3–4 hrs	Manageable	OPTIMAL for most athletes
5 meals/day	Excellent – near-continuous elevated MPS	Requires planning	Recommended for advanced athletes or high protein targets
6+ meals/day	No additional benefit over 5	Difficult	Not necessary. Can lead to GI issues.

Research consensus: 4–5 protein-containing meals per day optimally distribute MPS across the day.

Template: 160g Protein Goal (80kg Athlete @ 2g/kg)

MEAL	TIME	PROTEIN (g)	EXAMPLE FOODS
Meal 1 (Breakfast)	7:00–8:00 AM	35–40g	4 whole eggs + 2 egg whites + 300ml whole milk
Meal 2 (Lunch)	12:00–1:00 PM	40–45g	200g chicken breast + 1 cup dal + 200g rice
Meal 3 (Post-Workout Snack)	4:00–5:00 PM	25–30g	1 scoop whey protein + banana
Meal 4 (Dinner)	7:00–8:00 PM	35–40g	200g fish/paneer + 200g rice + vegetables
Meal 5 (Pre-Sleep)	9:30–10:00 PM	25–30g	200g paneer/cottage cheese OR 1 cup Greek yogurt
TOTAL	—	160–185g	Full protein distribution achieved across day

TIMING	MEAL SIZE	MACROS	BEST CHOICES
2–3 hrs before	Full meal (400–600 kcal)	Complex carbs 60–80g + Protein 30–40g + Low fat	Rice + chicken + vegetables, Oats + milk + eggs
60–90 min before	Light meal/snack (200–300 kcal)	Moderate carbs 40–60g + Protein 20–25g + Low fat	Banana + curd, Bread + egg whites, Small bowl of oats
30–45 min before	Small snack (100–200 kcal)	Simple carbs 30–40g + Light protein	Banana, Dates (3–5), Sports drink, Fruit
<30 min before	Minimal / liquid only	Simple carbs + caffeine	Coffee, pre-workout, banana

The Anabolic Window: 0–2 Hours Post-Training

PARAMETER	RECOMMENDATION	RATIONALE
Timing	Within 30–60 min of training end	Muscle is insulin-sensitive, MPS elevated for 24–48 hrs
Protein	30–50g fast protein (whey preferred)	Fast amino acids for MPS, leucine trigger

PARAMETER	RECOMMENDATION	RATIONALE
Carbohydrates	1–1.5 g/kg BW (60–120g)	Restore glycogen, spike insulin for anabolism
Fat	Keep low (<10g)	Fat slows amino acid absorption
Hydration	500–750ml water/sports drink	Replace fluid losses, restore electrolytes
Creatine (if using)	5g post-workout	Enhanced creatine uptake with insulin spike

Post-Workout Meal Examples by Profile

ATHLETE PROFILE	POST-WORKOUT MEAL EXAMPLE	APPROX PROTEIN	APPROX CARBS
Non-veg, gym access	1.5 scoop whey + banana + 200ml milk	45g	55g
Non-veg, no supplement	4 boiled eggs + 200g white rice + banana	36g	75g
Vegetarian, gym	1 scoop whey + 200ml milk + 50g oats cooked	35g	60g
Vegetarian, no supplement	1.5 cup paneer (150g) + 200g rice + apple	24g	70g
Hostel student	3 eggs + 2 slices bread + 1 glass milk + banana	30g	55g
Budget, non-veg	3 eggs + 200g rice + banana	25g	65g
Budget, vegetarian	100g soy chunks (soaked) + 200g rice + 300ml milk	44g	70g

- MPS occurs primarily during sleep (growth hormone peaks 1–2 hrs after sleep onset)
- Pre-sleep protein (30–40g casein) extends MPS throughout night by 7–8 hrs
- Casein (slow-digesting) is superior to whey for pre-sleep use
- Paneer, cottage cheese, Greek yogurt, milk are casein-rich whole food alternatives
- Pre-sleep carbs: moderate amount (50–100g) reduces cortisol, supports sleep quality
- Avoid high fat meals within 2 hrs of sleep – slows gastric emptying, disrupts sleep
- Avoid stimulants (caffeine) after 3 PM for athletes with sleep difficulties

FOOD	PROTEIN (g)	DIGESTION RATE	COST (■)	MPS DURATION
Casein Protein (supplement)	25g per scoop	Very slow (7 hrs)	60–100	7–8 hrs
Paneer (200g)	28g	Slow (casein-based)	50–100	6–7 hrs
Greek Yogurt (200g)	18–20g	Slow-moderate	60–100	5–6 hrs
Cottage Cheese (200g)	22g	Slow (casein-based)	40–80	6–7 hrs
Whole Milk (500ml)	17g	Mixed (whey + casein)	25–35	4–5 hrs
Whey protein (post-sleep caution)	25g per scoop	Very fast (1–2 hrs)	40–70	2–3 hrs only

SCENARIO	RECOMMENDATION	RATIONALE
Morning training, fasted	Take 10g EAAs or BCAA pre-workout	Minimizes muscle breakdown; EAAs trigger MPS
Training <1 hr after waking	Protein shake immediately upon waking	Breaks fast, minimizes catabolism
Fasted cardio (not weights)	Acceptable for low-intensity cardio only	Reduces muscle protein synthesis. Not ideal for weight training.
Ramadan / Fasting athlete	Shift training window to post-iftar (after breaking fast)	Train fed state. Eat within 30 min of training.
Intermittent Fasting (16:8)	Acceptable if protein targets met in 8hr window	Must achieve 4–5 meals of 35–40g protein in eating window

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- [R051] Minimum 4 protein meals per day. Each meal $\geq 30\text{g}$ protein for MPS stimulation.
 - [R052] Post-workout protein within 60 minutes. Whey protein is optimal. Whole food equivalent if no supplement.
 - [R053] Post-workout carbs: 1–1.5 g/kg BW within 2 hrs to maximize glycogen replenishment.
 - [R054] Pre-sleep protein mandatory for athletes seeking muscle gain. 30–40g paneer/casein/Greek yogurt.
 - [R055] Fasted weight training is suboptimal. Minimum 10g EAA/BCAA pre-training if fasted.
 - [R056] If athlete follows intermittent fasting: verify protein targets achievable in eating window before approving.
 - [R057] Pre-workout: no high-fat meal within 1.5 hrs of training. Fat slows gastric emptying.
 - [R058] Distribute protein evenly across meals. Avoid storing all protein in one meal.
 - [R059] Athletes with low appetite: liquid protein sources (shakes, milk) easier to consume post-training.
 - [R060] Meal timing is secondary to total daily intake. If athlete cannot eat within anabolic window, total daily protein still matters most.

MODULE 7

MUSCLE GAIN DECISION ENGINE

100+ IF-THEN coaching rules for all athlete types, budgets, lifestyles

DIMENSION	OPTIONS	IMPACT ON PLAN
Training Experience	Beginner (<1yr) Intermediate (1–3yr) Advanced (3+yr)	Volume, intensity, complexity of program
Diet Type	Non-vegetarian Vegetarian Vegan Egg-only	Protein sources, supplementation needs
Budget	Low (<₹150/day) Moderate (₹150–300/day) High (>₹300/day)	Food choices, supplementation
Living Situation	Home Hostel PG Independent (cooking)	Meal prep feasibility, food access
Schedule	Student (flexible) Working professional (limited time) Athlete (structured)	Meal frequency, training timing
Gym Access	Full gym Basic gym Home gym No gym	Exercise selection, equipment
Body Type	Ectomorph (lean, hard gainer) Mesomorph (muscular) Endomorph (gains fat easily)	Calorie surplus level
Body Fat	Lean (<10% M / <18% F) Normal Moderate High (>20% M / >28% F)	Surplus vs deficit vs recomp
Health Conditions	Healthy Lactose intolerant Diabetic GI issues Allergies	Food substitutions

IF athlete is VEGETARIAN (includes dairy + eggs) → THEN Base protein on eggs, dairy (paneer, milk, curd), soy chunks, lentils

IF athlete is VEGETARIAN AND protein target >150g/day → THEN Prioritize soy chunks, paneer, Greek yogurt, eggs, whey protein

IF athlete is VEGETARIAN AND high protein target AND low budget → THEN Soy chunks (52g/100g dry, ₹20), eggs, milk, roasted chana

IF athlete is VEGETARIAN AND lactose intolerant → THEN Use soy milk, tofu, soy chunks, eggs, pea protein. Avoid dairy.

IF athlete is VEGAN → THEN Soy protein isolate + pea protein blend, tofu, tempeh, lentils + rice combo. Supplement B12, D3, zinc, omega-3.

IF athlete is VEGETARIAN AND training heavy → THEN Protein must be ≥2.0 g/kg. Include soy at every meal.

IF athlete is VEGETARIAN AND cannot afford whey → THEN Soy chunks (dry) is most protein-dense budget option. 100g dry = 52g protein at ₹20–25.

IF athlete is VEGETARIAN AND eating 3 meals only → THEN Add protein-rich snack: roasted chana or peanuts between meals

IF athlete is VEGETARIAN AND eating eggs → THEN 4–6 eggs/day provides solid leucine and complete EAA profile

IF athlete is VEGETARIAN AND has dairy allergy → THEN Soy protein isolate is primary protein source. Add pea protein for variety.

IF hostel student AND low budget (₹100–150/day) → THEN Eggs (6/day) + milk (500ml) + soy chunks (100g dry) + dal + rice = ~130–150g protein at ₹80–120

IF hostel student AND access to mess/canteen → THEN Request: boiled eggs, dal, milk. Supplement with soy chunks in room.

IF hostel student AND no cooking facility → THEN Raw eggs (safe if fresh), peanut butter, roasted chana, canned tuna, protein powder + water

IF hostel student AND limited refrigeration → THEN Focus on shelf-stable proteins: soy chunks (dry), peanuts, chana, protein powder, canned fish

IF hostel student AND vegetarian → THEN Soy chunks + dal + milk + eggs (if lacto-ovo) = complete protein strategy

IF hostel student AND no gym membership → THEN Bodyweight training + resistance bands. Push-ups, pulls, squats, lunges, dips.

IF hostel student AND early morning class schedule → THEN Pre-workout: banana + peanut butter + milk at 6 AM. Post: eggs + bread + milk after class.

IF hostel student AND mess food only → THEN Eat dal + sabzi every meal. Add extra protein via eggs purchased separately. Supplement with whey if possible.

IF hostel student AND budget █50–80/day food → THEN Emergency protocol: 500g soy chunks powder (█200/month) + 30 eggs (█90) + milk packets. Minimum viable protein.

IF hostel student AND no time to cook → THEN Soaked soy chunks in cold water for 30 min, drain, season with salt/spice. Eat directly. 5 min prep.

IF working professional AND limited time for cooking → THEN Batch cook on weekends. 5-day meal prep: chicken/eggs + rice + vegetables.

IF working professional AND skipping breakfast → THEN Protein shake (30g whey + 300ml milk + banana) in 2 minutes.

Non-negotiable.

IF working professional AND office canteen available → THEN Order high-protein options: chicken salad, paneer dishes, dal. Skip rice if needed, not protein.

IF working professional AND training after work (evening) → THEN Pre-workout meal 1.5 hrs before training. Post-workout: whey shake + banana in locker.

IF working professional AND training in morning → THEN Wake 60 min early. Protein shake pre-workout. Full breakfast post-workout before office.

IF working professional AND high-stress job → THEN Cortisol management critical. Sleep 7+ hrs. Supplement: Ashwagandha 300–600mg. Reduce training volume temporarily.

IF working professional AND travel/business trips → THEN Pack protein powder single-serve sachets + nuts + protein bars. Hotel: egg breakfast. Order chicken/fish for dinner.

IF working professional AND meal service/tiffin available → THEN Request high-protein tiffin: 200g chicken + dal + 2 roti + vegetable. Supplement with eggs and milk.

IF working professional AND no evening energy for training → THEN Switch to morning training (24% more consistent long-term). Evening: walks, stretching.

IF working professional AND budget █400–600/day → THEN Adequate budget for chicken + fish + eggs + supplements. Lean bulk with full protein target achievable.

IF athlete has FULL GYM ACCESS (barbells, cables, machines) → THEN Use full progressive overload barbell program. Squat, deadlift, bench, row, OHP.

IF athlete has BASIC GYM (dumbbells, machines only) → THEN Dumbbell equivalents: DB press, DB row, goblet squat, Romanian DL, lateral raises.

IF athlete has HOME GYM with pull-up bar + bands only → THEN Push-up variations, pull-ups, band rows, resistance band squats. Effective for beginner-intermediate.

IF athlete has NO GYM and NO equipment → THEN Bodyweight program: progression via harder variations, tempos, pauses, unilateral work.

IF athlete training at HOME AND advanced → THEN Recommend weighted vest, dumbbells, or resistance bands purchase for continued overload.

IF athlete has GYM ACCESS but inconsistent attendance → THEN Create 3-day full-body program. Maximizes stimulus on training days available.

IF athlete has GYM ACCESS but no trainer → THEN Provide self-coached program with form cues. Emphasize video recording for form check.

IF athlete training at GYM with group classes only (Zumba, aerobics) → THEN Classes are insufficient for hypertrophy. Add 2–3 resistance training sessions/week minimum.

IF Male BF <8% OR Female BF <16% → THEN Aggressive bulk acceptable: TDEE +500 kcal. Higher surplus tolerated.

IF Male BF 8–15% OR Female BF 16–23% → THEN Lean bulk: TDEE +200–300 kcal. Sweet spot for muscle gain.

IF Male BF 15–20% OR Female BF 23–28% → THEN Conservative lean bulk: TDEE +150–200 kcal. Monitor fat gain carefully.

IF Male BF >20% OR Female BF >28% → THEN STOP surplus. Run recomp (TDEE ±0) or mini-cut (TDEE –250–300 kcal).

IF athlete GAINED 2+ kg in 4 weeks on bulk → THEN Excess fat gain. Reduce calories by 200–300 kcal. Maintain protein.

IF athlete NOT gaining weight on surplus for 3+ weeks → THEN Recalculate TDEE. Increase by +200 kcal. Check tracking accuracy.

IF athlete EXPERIENCING fat gain with very low surplus → THEN Check TDEE accuracy, NEAT reduction, water retention. Reassess.

IF athlete at maintenance BUT losing strength → THEN Insufficient calories AND/OR protein. Increase both. Check recovery.

BUDGET TIER	DAILY PROTEIN STRATEGY	CARB STRATEGY	FAT STRATEGY	SUPPLEMENT
Ultra Low (■80–120/day)	Eggs (6/day=■35) + Soy chunks 100g dry (■20) + Dal (■15) + Milk 500ml (■30) → ~120–130g protein	Rice (kg bag, cheapest bulk) + Chapati	Groundnut oil (cooking) + Peanuts 50g	None required
Low (■120–200/day)	Above + canned fish 100g (■40–60) 3x/week → ~140–150g protein	Rice + oats breakfast	Ghee 1 tsp + groundnut oil	Creatine only (■500/month)
Moderate (■200–350/day)	Eggs + chicken 200g/day (■60–80) + Dal + Milk → ~160–180g protein	Rice + oats + banana	Ghee + olive oil occasionally	Creatine + Whey 1 scoop
High (■350–600/day)	Chicken + eggs + fish + dairy → 180–200g+ protein	Brown rice + oats + fruits	Olive oil + avocado + fish oil	Whey + creatine + fish oil
Premium (■600+/day)	Any protein sources + supplements	Any complex carbs	Any healthy fats	Full stack: whey, creatine, fish oil, multivitamin

IF athlete is LACTOSE INTOLERANT → THEN Use lactose-free milk, hard cheese (aged), soy milk, tofu, pea/soy protein. Avoid whey concentrate (use isolate or plant protein).

IF athlete reports DIGESTIVE ISSUES with protein shakes → THEN Switch from whey concentrate to whey isolate. Reduce powder quantity. Space doses.

IF athlete has EGG ALLERGY → THEN Replace eggs with: soy chunks, paneer, legumes. Supplement with pea/soy protein.

IF athlete is DIABETIC (Type 2) → THEN Lower carb diet 150–200g/day. Higher protein 2.0–2.4g/kg. Complex carbs only. Frequent small meals.

IF athlete has HYPERTENSION → THEN Reduce sodium. Avoid canned/processed foods. Fresh whole foods. Check creatine (generally safe but monitor).

IF athlete has PCOS (female) → THEN Lower GI foods. Moderate carbs. Higher protein 1.8–2.0g/kg. Resistance training especially beneficial.

IF athlete is UNDERWEIGHT (BMI <17) → THEN Moderate aggressive bulk +400–500 kcal. High calorie dense foods. Liquid calories help.

IF athlete is RECOVERING FROM INJURY → THEN Maintain protein 2.0–2.2 g/kg. Reduce training volume. Prioritize sleep. No deficit.

IF athlete has POOR SLEEP HISTORY → THEN Address sleep before increasing training. Reduce volume 20%. Add magnesium glycinate 300–400mg night.

IF athlete is UNDER 18 (minor) → THEN Focus on compound movements, technique. Avoid excessive calorie surplus. No supplements beyond food.

- [R061] ALWAYS classify athlete on 9 dimensions before generating any plan.
- [R062] Diet type (veg/non-veg) is the most critical variable for protein source selection.
- [R063] Budget is the most limiting practical constraint. Design within actual budget.
- [R064] Hostel students: soy chunks + eggs + milk = minimum viable protein stack.
- [R065] Working professionals: batch cooking + liquid protein = compliance strategy.
- [R066] No gym access is NOT an excuse for no training. Bodyweight program is effective for beginners.
- [R067] Health conditions must be flagged before supplement or nutrition recommendations.
- [R068] Female athletes: same protein principles apply relative to bodyweight. Monthly cycle affects recovery.
- [R069] Athletes with BF >20% (M) or >28% (F): run recomp before bulk. Do not add surplus.
- [R070] Lactose intolerant athletes: whey isolate is tolerated by most. Recommend if needed.

MODULE 8

EXERCISE SCIENCE

Training volume, frequency, intensity, compound/isolation, splits, and key lift coaching

VARIABLE	DEFINITION	UNIT	HYPERTROPHY OPTIMAL RANGE
Volume	Total work performed = Sets × Reps × Load	Sets per muscle/week	10–20 sets/muscle/week
Intensity	Load relative to 1 rep max	% of 1RM	60–85% 1RM
Frequency	How often a muscle is trained per week	Sessions/muscle/week	2–4x/week
Rep Range	Number of reps per set	Reps per set	6–12 primary, 15–20 secondary
Rest Interval	Rest between sets	Seconds / Minutes	60–180 seconds
Tempo	Speed of movement phases (eccentric-pause-concentric)	X-X-X format	2-1-2 to 3-1-1
Time Under Tension (TUT)	Total time muscle is under load per set	Seconds per set	30–60 seconds
RPE (Rate of Perceived Exertion)	Subjective intensity scale	0–10 (10 = max)	RPE 7–9 for most work sets
RIR (Reps in Reserve)	Reps left in tank at set completion	Reps	1–3 RIR optimal

MUSCLE GROUP	MINIMUM EFFECTIVE VOLUME	OPTIMAL VOLUME	MAXIMUM RECOVERABLE VOLUME	NOTES
Chest	8 sets/week	12–16 sets/week	20–22 sets/week	Include incline + flat + fly movements
Back (all)	10 sets/week	14–20 sets/week	25 sets/week	Vertical + horizontal pull
Shoulders	8 sets/week	12–16 sets/week	20 sets/week	All 3 heads. Lateral raises important.
Biceps	6 sets/week	10–14 sets/week	20 sets/week	Often trained via back work
Triceps	6 sets/week	10–14 sets/week	18 sets/week	Often trained via chest/shoulder work
Quads	8 sets/week	12–16 sets/week	20 sets/week	Squat + leg press + extensions
Hamstrings	6 sets/week	10–14 sets/week	18 sets/week	Deadlifts + curls
Glutes	4 sets/week	8–14 sets/week	20 sets/week	Hip thrusts, RDL, squat
Calves	8 sets/week	12–16 sets/week	20 sets/week	High frequency responds well
Core (Abs)	6 sets/week	10–14 sets/week	16 sets/week	Compound lifts provide significant stimulus
Forearms	2–4 sets/week	6–8 sets/week	12 sets/week	Mostly trained via pulling

MV = Minimum Effective Volume. MRV = Maximum Recoverable Volume. Start near MV and progress over program.

REP RANGE	LOAD (%1RM)	PRIMARY ADAPTATION	HYPERTROPHY POTENTIAL	REST PERIOD
1–3 reps	95–100%	Max strength, neural efficiency	Low (insufficient TUT)	3–5 min

REP RANGE	LOAD (%1RM)	PRIMARY ADAPTATION	HYPERTROPHY POTENTIAL	REST PERIOD
4–6 reps	85–95%	Strength + myofibrillar hypertrophy	Moderate-High	2–4 min
6–12 reps	67–85%	Myofibrillar + sarcoplasmic hypertrophy	HIGH (sweet spot)	1–3 min
12–20 reps	50–67%	Sarcoplasmic hypertrophy + metabolic stress	Moderate-High	60–90 sec
20–30 reps	40–50%	Metabolic stress + muscular endurance	Low-Moderate	30–60 sec
30+ reps	<40%	Muscular endurance	Low	30 sec

All rep ranges produce hypertrophy when taken to near failure (RIR 1–3). 6–12 is optimal for most. Include variety for complete development.

SPLIT	FREQUENCY	SESSIONS/ WEEK	BEST FOR	EXAMPLE SCHEDULE
Full Body	Each muscle 3x/week	3	Beginners, limited time, 3 days/week	Mon: Full / Wed: Full / Fri: Full
Upper/Lower	Each muscle 2x/week	4	Intermediate, balanced, 4 days/week	Mon: Upper / Tue: Lower / Thu: Upper / Fri: Lower
Push/Pull/Legs (PPL)	Each muscle 2x/week	6	Intermediate-Advanced, 6 days	Mon:Push / Tue:Pull / Wed:Legs / Thu:Push / Fri:Pull / Sat:Legs
PPL 3-day version	Each muscle 1x/week	3	Beginners to intermediate	Mon:Push / Wed:Pull / Fri:Legs
Bro Split	Each muscle 1x/week	5	Advanced bodybuilders only	Chest/Back/Shoulders/Arms/Legs
Arnold Split	Each muscle 2x/week	6	Advanced, high volume	Chest+Back/Shoulders+Arms/Legs (×2)
2-Day Split	Each muscle 1–2x/week	2	Very busy schedules	Full body or Upper/Lower

TYPE	DEFINITION	BENEFITS	EXAMPLES	PLACEMENT IN WORKOUT
Compound (Multi-joint)	Multiple muscle groups, multiple joints. High neural demand.	High anabolic stimulus, efficient time use, hormonal response	Squat, Deadlift, Bench, Row, OHP, Pull-up	First in session. When fresh.
Isolation (Single-joint)	One muscle, one joint. Lower neural demand.	Targeted hypertrophy, address weaknesses, pump	Curls, extensions, lateral raises, flyes, leg curls	After compound work.

Ratio recommendation: 60–70% compound movements, 30–40% isolation. Beginners should use 80–90% compound.

SQUAT (Back Squat)

ATTRIBUTE	DETAIL
Primary Muscles	Quads (primary), Glutes, Hamstrings, Spinal Erectors, Core
Setup	Bar on traps (high bar) or delts (low bar). Feet shoulder-width+, toes out 15–30°.
Coaching Cues	Brace core: 360° tension (Valsalva maneuver for heavy sets) Initiate: hip hinge + knee push simultaneously Depth: crease of hip below knee (at minimum parallel) Knees track over toes. No knee cave. Drive floor away. Chest up on way up.
Progressions/Variants	Goblet squat → Box squat → Front squat → Bulgarian split squat → Leg press
Load Progression	Add 2.5 kg per session beginners, 2.5 kg per week intermediates

DEADLIFT (Conventional)

ATTRIBUTE	DETAIL
Primary Muscles	Glutes + Hamstrings (primary), Erectors, Lats, Traps, Core
Setup	Bar over midfoot, hip-width stance. Hip hinge, grab bar just outside legs.
Coaching Cues	Lat engagement: 'bend the bar' or 'protect armpits' cue Brace before lift – maintain through entire rep Drive: leg press the floor → hips forward at lock Bar stays in contact with body throughout pull Hinge at hip to lower (not bend the back)
Progressions/Variants	Romanian DL → Trap bar DL → Sumo DL → Good mornings → Hip thrust
Load Progression	Add 2.5 kg per session beginners, 5 kg per week intermediates

BENCH PRESS (Flat Barbell)

ATTRIBUTE	DETAIL
Primary Muscles	Pectorals (primary), Anterior Deltoid, Triceps
Setup	Eyes under bar. 5 points: head, upper back, glutes, both feet on floor. Grip 1.5–2x shoulder width.
Coaching Cues	Arch upper back (not lumbar). Scapulae retracted + depressed. Unrack to directly over chest. Control descent. Touch lower chest (below nipple line). No bounce. Drive elbows slightly in (45°). Press and slightly arc. Push feet into floor. Leg drive transfers through glutes to press.
Progressions/Variants	DB Press → Incline DB Press → Dips → Push-ups → Cable Fly
Load Progression	Add 2.5 kg per session beginners, 2.5 kg per week intermediates

PULL-UP / CHIN-UP

ATTRIBUTE	DETAIL
Primary Muscles	Latissimus Dorsi (primary), Biceps, Rear Delts, Rhomboids
Setup	Overhand (pull-up) or underhand (chin-up). Shoulder-width or wider grip. Full hang start.
Coaching Cues	Initiate with shoulder depression (pack scapulae down) Pull chest to bar – not chin Avoid kipping or swinging for hypertrophy Control lowering – eccentric is critical for hypertrophy Full extension at bottom (don't shortchange stretch)
Progressions/Variants	Assisted pull-up → Band-assisted → Lat pulldown → Neutral grip → Weighted pull-up
Load Progression	1 extra rep per week. Add weight when 3×12 achieved.

BARBELL ROW (Bent-Over Row)

ATTRIBUTE	DETAIL
Primary Muscles	Lats, Rhomboids, Traps, Rear Delts, Biceps
Setup	Hip hinge, torso 30–45° from floor. Overhand grip, shoulder-width.
Coaching Cues	Pull to lower chest/upper abdomen (not chin) Lead with elbows, not hands Retract scapulae at top – squeeze between shoulder blades Maintain tight brace throughout set Avoid jerking. Controlled tempo.
Progressions/Variants	DB Row → Chest-supported row → Cable row → T-bar row → Pendlay row
Load Progression	Add 2.5 kg per week. DB row progression same.

OVERHEAD PRESS (Barbell/Dumbbell)

ATTRIBUTE	DETAIL
Primary Muscles	Anterior + Medial Deltoid (primary), Triceps, Upper Traps
Setup	Bar in hands at shoulder height, elbows slightly forward. Tight grip. Core braced.
Coaching Cues	Press straight up – slight backward lean at lockout, not forward Head moves back briefly to clear path, then forward at lockout Squeeze glutes + brace abs to prevent lumbar extension Full lockout at top. No elbow flare in initial drive. Control descent to shoulder level. Pause. Press again.
Progressions/Variants	DB OHP → Seated OHP → Landmine press → Arnold press → Machine shoulder press
Load Progression	Add 1.25–2.5 kg per week for beginners

GOAL	REST INTERVAL	RATIONALE	EXAMPLE APPLICATION
Max Strength	3–5 min	Full PCr restoration, full CNS recovery	Compound lifts 1–5 RM
Hypertrophy (primary)	1.5–3 min	Partial recovery + metabolic stress maintained	Compound 6–12 RM
Hypertrophy (pump/metabolic)	60–90 sec	High metabolic stress, pump, sarcoplasmic	Isolation, supersets, drop sets
Muscular Endurance	30–60 sec	Lactate tolerance, cardiovascular	Circuit training
Supersets (antagonist)	0 sec between pair, 90 sec after pair	Efficient time use, agonist recovery while antagonist works	Bicep curl + tricep ext
Dropsets	0 sec during drops, 2–3 min after	Maximum metabolic stress in single set	Final set of isolation exercises

MOVEMENT	BEGINNER	INTERMEDIATE	ADVANCED
Push (Chest/Tris)	Wall push-up → Incline push-up	Push-up → Diamond push-up	Decline push-up → Pike push-up → Handstand push-up
Pull (Back/Bis)	Bodyweight row (table) → Band pull-apart	Chin-up (assisted) → Pull-up	Weighted pull-up → Muscle-up
Squat (Quads/Glutes)	Squat to chair → Air squat	Squat → Jump squat	Bulgarian split squat → Pistol squat
Hip Hinge (Hamstrings/Glutes)	Glute bridge → Hip thrust	Single-leg glute bridge → Nordic curl	Shrimp squat → Nordic hamstring curl
Core	Dead bug → Hollow hold	Plank → Ab wheel	Dragon flag → Hanging leg raise
Shoulder	Pike push-up → Band lateral raise	Handstand hold → DB raise (if available)	Handstand push-up → Weighted

- [R071] Beginners: 3-day full body program with compound movements only. No isolation for first 3–6 months.
- [R072] Progressive overload is mandatory in every session. Track all lifts. Record weight, sets, reps.
- [R073] Rep range 6–12 is the primary hypertrophy zone. Maintain variety of 4–6 heavy and 12–20 light ranges.
- [R074] Compound movements first in every session. Isolation movements after.
- [R075] Beginner: add weight every session. Intermediate: add weight every week. Advanced: add weight every mesocycle.
- [R076] Minimum 2x/week frequency per muscle group for intermediate-advanced athletes.
- [R077] Volume: start at 10 sets/muscle/week. Increase by 2 sets per 4-week block.
- [R078] Rest: 2–3 min for compound heavy work. 60–90 sec for isolation/pump work.
- [R079] Bodyweight athletes: progression through harder variations. No equipment = no excuse.
- [R080] Form over weight, always. Flag form breakdown as injury risk. Reduce load, fix form, reload.

MODULE 9

RECOVERY SCIENCE

Sleep, stress, deloads, CNS fatigue, overtraining, and active recovery

PARAMETER	RECOMMENDATION	CONSEQUENCE OF DEFICIENCY
Duration	7–9 hours per night (minimum 7)	1 week at 5hr sleep = 15% testosterone drop, impaired MPS
Sleep Quality	Uninterrupted REM + deep sleep cycles	Fragmented sleep reduces GH release and protein synthesis
Consistency	Same sleep/wake time daily (±30 min)	Irregular sleep disrupts cortisol rhythm and anabolic hormones
Pre-sleep environment	Dark, cool (18–20°C), quiet	Bright light and heat impair melatonin and deep sleep
Pre-sleep nutrition	30–40g casein protein 30–60 min before bed	Enhances overnight MPS by 22% (Res 2012)
Alcohol	AVOID completely	Even 1 drink impairs GH release by 70% in first sleep cycle
Caffeine	No caffeine after 2–3 PM for 10hr half-life effects	Caffeine impairs sleep latency and quality
Naps	20–30 min naps acceptable	90+ min naps reduce nighttime sleep quality

- GH is released in pulses, primarily during deep sleep (slow-wave sleep stages 3–4)
- Peak GH pulse occurs 1–2 hours after sleep onset
- GH stimulates protein synthesis, fat mobilization, and recovery
- Sleep deprivation reduces GH pulse amplitude by 50–70%
- Alcohol suppresses GH secretion significantly in first sleep cycle
- High blood glucose before sleep blunts GH secretion (insulin inhibits GH)
- Moderate carbs + casein protein pre-sleep is optimal – not high-sugar foods

CORTISOL TRIGGER	EFFECT ON MUSCLE	MANAGEMENT STRATEGY
Chronic high stress (job, exam, life)	Cortisol is catabolic – breaks down muscle protein, inhibits testosterone	Reduce training volume during high-stress periods
Overtraining	Chronically elevated cortisol, reduced anabolic hormones	Deload, increase sleep, reduce training frequency
Sleep deprivation	Cortisol rises with poor sleep, GH drops	Prioritize sleep over extra training
Caloric deficit	Deficit > 500 kcal elevates cortisol significantly	Stay within lean bulk range, avoid crash dieting
Fasted training	Elevated cortisol + reduced insulin during training	Take EAAs/BCAAs before fasted training
Caffeine overuse	Excess caffeine chronically elevates cortisol	Limit to 200–400 mg/day. Cycle off monthly.
Social/emotional stress	Chronic stress impairs recovery despite adequate sleep	Mindfulness, physical activity variety, social support

RECOVERY METHOD	EVIDENCE LEVEL	MECHANISM	PRACTICAL APPLICATION
Sleep (7–9 hrs)	Strong	GH release, protein synthesis, cortisol reduction	Non-negotiable. First recovery priority.
Nutrition (protein + carbs post-workout)	Strong	Glycogen replenishment, MPS stimulation	30–50g protein + 60–120g carbs within 2 hrs
Active recovery (low-intensity movement)	Moderate	Blood flow, waste removal, reduces DOMS	Walk, swim, yoga, light cycling on rest days
Cold water immersion (CWI)	Moderate	Reduces acute inflammation and DOMS	Cold shower or ice bath 10–15 min post-training
Massage / foam rolling	Moderate	Increases blood flow, reduces soreness perception	10–15 min foam rolling post-workout
Contrast therapy (hot/cold)	Moderate	Vasodilation/vasoconstriction cycling	3 min hot, 1 min cold, repeat 3–4 cycles
Sauna (dry or infrared)	Moderate	Heat shock proteins, GH boost, relaxation	15–20 min sauna session post-workout
Compression garments	Low-Moderate	Reduce swelling, improve venous return	Wear post-exercise for 24–48 hrs
Stretching (static, post-workout)	Low	Increases ROM, may marginally reduce DOMS	5–10 min static stretching after training
Deload	Strong	Reduces accumulated fatigue, allows super-compensation	1 week every 4–6 weeks of hard training

When to Deload

- Every 4–6 weeks of hard training (scheduled deload)
- If joints are chronically sore or achy
- If strength has declined for 2+ consecutive weeks
- If sleep is disrupted and recovery feels incomplete
- If mood, motivation, and energy are persistently low
- If training sessions are consistently poor quality

DELOAD TYPE	VOLUME CHANGE	INTENSITY CHANGE	FREQUENCY CHANGE	BEST FOR
Volume Deload (preferred)	Reduce by 50%	Maintain (70–85% 1RM)	Maintain	Most athletes. Maintain neural drive.
Intensity Deload	Maintain	Reduce to 50–60% 1RM	Maintain	Joint pain, technique refinement
Full Deload (rest week)	0 (rest or walk)	0	0	Severe fatigue, illness, overtraining
Frequency Deload	Reduce by 30%	Maintain	Reduce by 50%	Busy schedule, moderate fatigue

Overtraining Syndrome Indicators

CATEGORY	SYMPTOMS	SEVERITY
Performance	Strength decline >10% over 2+ weeks, slower times, reduced power	HIGH CONCERN
Physical	Persistent fatigue, DOMS lasting >72 hrs, frequent illness, elevated resting HR	HIGH CONCERN
Hormonal	Testosterone/Cortisol ratio decreases. Libido reduction. Irregular menstrual cycle (females)	HIGH CONCERN
Psychological	Reduced motivation, mood swings, depression, irritability, poor concentration	MODERATE CONCERN
Sleep	Insomnia despite fatigue, restless sleep, excessive sleeping	MODERATE CONCERN
Appetite	Loss of appetite, weight loss despite adequate calories, nausea	MODERATE CONCERN

OTS Management Protocol

- Phase 1 (1–2 weeks): Complete rest from structured training. Sleep 9+ hrs. Eat at maintenance.
- Phase 2 (1–2 weeks): Light activity only (walking, yoga). Continue good nutrition.
- Phase 3 (1–2 weeks): Resume training at 50% of previous volume. Exclude heavy compound lifts.
- Phase 4: Gradually return to normal program over 4–6 weeks.
- Prevention: Build deloads into program. Monitor training logs for early warning signs.

- CNS fatigue is distinct from muscular fatigue. Muscles may feel fine but neural drive is impaired.
- Indicators: reduced bar speed, grinding reps despite no strength loss, low motivation
- Heavy compound work (deadlifts, squats) is most CNS demanding
- Recovery from CNS fatigue: 48–72 hrs minimum for heavy sessions
- Strategy: alternate CNS-demanding (heavy) sessions with CNS-lighter (isolation/volume) sessions
- Caffeine can temporarily mask CNS fatigue – do not use as long-term solution
- Nutrition support: carbohydrates restore glycogen + support neurotransmitter function

[R081] Sleep is the first recovery priority. Below 7 hours consistently = flag as high-risk for muscle loss.

[R082] Schedule deloads every 4–6 weeks. Volume deload is preferred method.

[R083] If strength declines for 2 consecutive weeks without explanation, prescribe deload immediately.

[R084] Chronic stress (life/work) must reduce training volume to compensate. Do not increase volume during high-stress periods.

[R085] Pre-sleep nutrition: 30–40g casein protein recommended for all bulking athletes.

[R086] OTS (overtraining syndrome): prescribe complete rest. Not a deload. Full stop from training 1–2 weeks.

[R087] Alcohol consumption impairs GH, testosterone, and sleep quality. Zero tolerance during bulk phases.

[R088] Active recovery on rest days improves blood flow and reduces DOMS without taxing CNS.

[R089] Foam rolling + static stretching post-workout: recommend 10–15 min for injury prevention.

[R090] Recovery formula: Adequate Sleep + Calorie Surplus + Low Stress + Deload = Optimal Recovery.

MODULE 10

SUPPLEMENTS

Evidence-based only. No steroids, no illegal substances.

TIER	DEFINITION	SUPPLEMENTS
Tier 1 – Strong Evidence	Consistent, replicated evidence in humans. Recommend freely.	Creatine monohydrate, Whey protein, Caffeine
Tier 2 – Moderate Evidence	Good evidence but some conflicting studies. Recommend with context.	Fish oil (omega-3), Vitamin D, Magnesium
Tier 3 – Emerging Evidence	Early or limited human evidence. Use with caution.	Ashwagandha, HMB (for older adults), Citrulline
Tier 4 – Weak/No Evidence	Mostly marketing. Do not recommend.	BCAAs (if protein met), Glutamine, Testosterone boosters, Fat burners
Banned/Illegal	Steroids, SARMs, EPO, HGH, clenbuterol, prohormones	NEVER recommend. Flag if athlete mentions.

Most Researched, Most Effective Legal Supplement for Muscle Gain

PARAMETER	DETAIL
PARAMETER	DETAIL
Mechanism	Replenishes PCr (phosphocreatine) stores → more ATP available for high-intensity work → higher training volume → more muscle gain
Effect on muscle gain	Meta-analyses: +1.2–2.5 kg lean mass increase over 4–12 week programs vs placebo
Effect on strength	Strength gains 5–15% greater than training alone
Dose (standard)	3–5g per day. No loading phase needed (loading = faster saturation but same end result in 28 days)
Dose (loading optional)	20g/day in 4 doses × 5 days, then 3–5g/day maintenance
Timing	Post-workout is marginally better. Time of day matters less than consistency.
Form	Creatine MONOHYDRATE only. No need for ethyl ester, buffered, kre-alkalyn. Marketing gimmicks.
Side effects	Water retention (intramuscular, not subcutaneous) – normal and desirable. Safe for kidneys in healthy individuals.
Contraindications	Pre-existing kidney disease: medical clearance required. Avoid in kidney dysfunction.
Cost (India)	■400–800 per month for branded. Generic monohydrate widely available.
Evidence level	TIER 1. 1000+ studies. Strongest evidence of any sports supplement.

PARAMETER	DETAIL
PARAMETER	DETAIL
Source	By-product of cheese production. Liquid whey filtered and dried.
Types	Concentrate (70–80% protein), Isolate (90%+ protein, less lactose), Hydrolysate (pre-digested, fastest)
Best type for most	Whey Concentrate (cost-effective). Isolate for lactose intolerant.
Protein per serving	24–26g per 30g scoop (varies by brand)

PARAMETER	DETAIL
Leucine content	~2.7g per 30g scoop (best of all protein sources)
Absorption rate	Fast: 8–10g per hour. Reaches bloodstream in 30–60 min.
Best use	Post-workout (fast absorption). Morning (break fast quickly).
Is it necessary?	NO – whole foods can replace it. Supplement only if dietary protein targets not met from food.
Cost (India)	■1,000–3,000 per kg. 30–50 servings. ■30–100 per serving.
Quality markers	Look for: protein per 100g >70g, labdoor-tested brands, no prop blends, no soy filler
Recommended brands (India)	MuscleBlaze, Optimum Nutrition, BigMuscles Nutrition, Dymatize (imported)

PARAMETER	DETAIL
PARAMETER	DETAIL
Active components	EPA (eicosapentaenoic acid) + DHA (docosahexaenoic acid)
Benefits for muscle gain	Reduces muscle soreness, enhances MPS response to protein, improves insulin sensitivity, reduces inflammation
Dose	2–4g combined EPA+DHA per day (not fish oil total – check EPA+DHA on label)
Timing	With meals. Fat-soluble. Better absorbed with dietary fat.
Natural food sources	Sardines (3.5g/100g EPA+DHA), Salmon (2.5g/100g), Mackerel (2.5g/100g), Tuna (0.5g/100g)
For vegetarians/vegans	Algae oil (direct EPA+DHA source, not reliant on fish). Flaxseed provides ALA only (poorly converted).
Safety	Generally very safe. High doses >4g/day may have anticoagulant effects.
Cost (India)	■300–800 per month for quality supplement. Budget brands available.
Quality markers	IFOS-certified, triglyceride form preferred over ethyl ester, no fishy smell/taste
Evidence level	TIER 2. Strong anti-inflammatory evidence. Muscle gain evidence emerging.

PARAMETER	DETAIL
PARAMETER	DETAIL
Purpose	Insurance against micronutrient deficiencies. NOT a replacement for whole food diet.
Key nutrients for athletes	Vitamin D (1000–2000 IU), Magnesium (300–400 mg), Zinc (10–15 mg), B-complex, Iron (females)
Who needs it most	Athletes with restricted diets (vegetarian, calorie cutting), hostel students with poor food variety
Iron (females)	Female athletes at high risk of iron deficiency anemia. Check ferritin levels. Supplement if deficient.
Vitamin D	Majority of Indian population deficient. Critical for testosterone, MPS, bone health. Supplement if not getting sun.
Zinc	Lost in sweat. Athletes at higher risk of deficiency. 11mg/day men, 8mg/day women.
Dosing	One multivitamin daily with largest meal. Do not exceed upper tolerable limits.
Evidence for muscle gain	TIER 3 for direct muscle gain. TIER 1 for correcting deficiency (which impairs performance).
Cost (India)	■200–1,000 per month

PARAMETER	DETAIL
PARAMETER	DETAIL
Mechanism	Adenosine receptor antagonist. Reduces perceived effort, increases CNS activation, improves strength output by 3–7%.

PARAMETER	DETAIL
Benefits	Strength +3–7%, endurance, delayed fatigue, mental focus, thermogenesis
Dose	3–6 mg/kg bodyweight (200–400 mg for most). Start low to assess tolerance.
Timing	30–60 min before training. Half-life: 5–7 hrs. No caffeine after 2–3 PM.
Sources	Coffee (80–120mg/cup), Black tea (40–70mg), Pre-workout supplements, Caffeine pills (200mg)
Tolerance	Develops within 1–2 weeks. Cycle off 1–2 weeks/month to reset sensitivity.
Side effects	Anxiety, tachycardia, insomnia at high doses. GI issues in sensitive individuals.
Contraindications	Hypertension, anxiety disorder, heart arrhythmia. Medical clearance recommended.
Evidence level	TIER 1. Performance enhancement well-established.

ELECTROLYTE	FUNCTION	DEFICIENCY SIGNS	SOURCES	ATHLETE DOSE
Sodium	Fluid balance, nerve function, muscle contraction	Cramping, fatigue, low performance	Salt, sodium chloride, pickles	1500–3000 mg/day (higher in heat/heavy sweaters)
Potassium	Muscle contraction, fluid balance	Cramping, weakness	Banana, potato, coconut water, dal	3500–4700 mg/day
Magnesium	Protein synthesis, energy (ATP), muscle relaxation	Cramping, poor sleep, low testosterone	Dark leafy greens, nuts, legumes	400–420 mg/day (men), 310–360 (women)
Calcium	Muscle contraction, bone health	Cramping, bone injury	Dairy, fortified milk, ragi	1000–1200 mg/day
Chloride	Fluid balance, stomach acid	Rare in isolation	Salt	From sodium chloride intake

PROFILE	PRIORITY 1	PRIORITY 2	PRIORITY 3	OPTIONAL
Budget athlete (any)	Good food first	Creatine (■400–500/mo)	—	Multivitamin if diet poor
Vegetarian + low protein	Whey protein	Creatine	Vitamin B12 + D3	Fish oil (algae) + Zinc
Non-veg, gym 4x/week	Creatine	Whey (if needed)	Fish oil	Multivitamin + Caffeine
Hostel student (budget)	Creatine only if >■500 available	Multivitamin	—	—
Working professional	Creatine	Whey (convenience)	Fish oil	Vitamin D + Caffeine
Female athlete	Whey (if protein low)	Creatine	Iron + Vitamin D	Fish oil + Magnesium
Advanced athlete, bulking	Creatine + Whey	Fish oil	Vitamin D + Magnesium	Ashwagandha (stress)

[R091] Supplements are optional – food is mandatory. Never recommend supplement before dietary foundation is solid.

[R092] Creatine monohydrate is the #1 supplement recommendation for all resistance training athletes.

[R093] Whey protein: recommend only when athlete cannot meet protein targets from whole food.

[R094] Never recommend BCAAs if athlete is meeting total daily protein target. Money waste.

[R095] Fish oil: recommend if athlete is not eating fatty fish 2–3x/week.

[R096] Vitamin D3: supplement routinely for Indian athletes who spend most time indoors.

[R097] NEVER recommend steroids, SARMs, HGH, testosterone boosters claiming 'natural'. Flag and educate.

[R098] Caffeine: effective, but cycle to maintain sensitivity. No caffeine for sleep-compromised athletes.

[R099] For athletes under 18: food only. No supplements required or recommended.

[R100] Supplement budget priority: Creatine > Whey Protein > Fish Oil > Vitamin D > Multivitamin > Others.

MODULE 11

SPECIAL POPULATIONS

Skinny beginners, hardgainers, women, vegetarians, hostel students, busy professionals

Profile: BMI <18.5, Low muscle mass, Fast metabolism, Hard to gain weight

AREA	RECOMMENDATION
AREA	RECOMMENDATION
Calorie Target	TDEE + 400–600 kcal. Aggressive lean bulk. Start high.
Protein	1.8–2.0 g/kg. Distribute across 5 meals.
Carbs	5–7 g/kg. Load on carbs. White rice, oats, banana, dates.
Fats	0.8–1.2 g/kg. Include calorie-dense fats: peanut butter, ghee, nuts.
Liquid calories	Use milk (500ml = 300 kcal), fruit juice, smoothies to increase caloric intake with low volume.
Meal frequency	5–6 meals per day. Eat every 2.5–3 hrs. Set alarms if needed.
Training	3–4 day program. Compound movements primary. No cardio in early phases.
Sleep	8–9 hrs. Critical for GH release and weight gain.
Common mistake	Underestimating calories. Must track every meal. Real deficit hiding in poor counting.
Progress target	0.5–1.0 kg/week. Skinny beginners can gain faster than other groups.

Profile: Has been eating 'a lot' but not gaining – Usually underestimating intake

STEP	ACTION
STEP	ACTION
1. Audit actual intake	Make athlete log EVERY bite for 7 days via MyFitnessPal or paper. Most discover they eat far less than perceived.
2. Recalculate TDEE	Higher NEAT multiplier. Some athletes burn 3000+ kcal/day.
3. Increase calories +200 kcal	Every 2 weeks if not gaining. Keep adding until 0.25 kg/week gain achieved.
4. Add liquid calories	500ml whole milk + peanut butter shake = +600 kcal added to diet easily.
5. Reduce NEAT	Hardgainers often unconsciously fidget, move more. Reduce non-essential walking/activity.
6. Prioritize sleep	8–9 hrs minimum. Hardgainers often sleep less.
7. Reduce cardio	Eliminate all cardio in initial gain phase. Add back after 3–4 months if fat gain acceptable.
8. Consistency audit	Are they actually eating the target daily? Missing weekends? Skipping post-workout meal?

Key Differences vs Male Athletes

PARAMETER	FEMALE	COACHING NOTE
Testosterone level	~5–10% of male levels	Primary reason for slower absolute muscle gain
Muscle gain rate	40–60% of male rate	Realistic expectation setting critical
Estrogen role	Anabolic in muscle, anti-inflammatory	Beneficial for recovery, especially pre-menstrual phase
Menstrual cycle phases	Follicular (day 1–14): higher estrogen – strength peaks. Luteal (day 15–28): higher progesterone – fatigue, heat	Program heavier sessions in follicular phase
Protein target	1.6–2.0 g/kg (same as males relative to BW)	Often undereat protein. Emphasize.
Calorie target	Smaller surplus: +150–250 kcal typically sufficient	Lower TDEE than males. Less surplus needed.
Iron needs	Higher than males (18mg vs 8mg/day)	Vegetarian/vegan females at very high risk of deficiency
Body fat	Essential fat = 10–13% (females) vs 2–5% (males)	Do not push below 15% BF. Health risk.
Bone health	Higher osteoporosis risk	Resistance training beneficial for bone density
Strength goals	Different benchmarks	Female intermediate: Squat 1×BW, Deadlift 1.25×BW, Bench 0.5×BW
Training myths	Lifting heavy makes women 'bulky'	FALSE. Testosterone levels prevent excess bulk. Heavy lifting is safe and beneficial.

Protein Combination Table for Complete Amino Acid Profiles

COMBINATION	COMPLETE?	EXAMPLE MEAL	PROTEIN (approx)
Rice + Lentils (Dal)	YES	200g rice + 200g cooked dal	25–30g
Roti + Dal	YES	3 roti + 200g dal	22–28g
Milk + Wheat (Oats + Milk)	YES	100g oats + 300ml milk	22g
Soy chunks (complete alone)	YES	100g soy chunks dry + any side	52g
Eggs + anything	YES (eggs complete)	3 eggs + roti	25g+
Paneer + roti	Adequate	200g paneer + 2 roti	30g
Corn + Beans	YES	Corn + rajma salad	20–25g
Pea protein + Rice protein	YES	Protein blend supplement	Per serving

Vegetarian Athlete Daily Template (75kg Male, Protein Goal: 150g)

MEAL	FOODS	PROTEIN (g)
Breakfast (8 AM)	4 whole eggs + 300ml whole milk + 50g oats cooked	42g
Lunch (1 PM)	150g paneer (cooked) + 200g rice + 200g dal + vegetables	40g
Post-Workout Snack (5 PM)	1 scoop whey + 300ml milk + banana	38g
Dinner (8 PM)	100g soy chunks (cooked, ~52g protein dry) + 2 roti + sabzi	36g
Pre-Sleep (10 PM)	200g Greek yogurt / curd + roasted chana 50g	20g
TOTAL	—	176g

BUDGET	PROTEIN PLAN	CARB PLAN	FAT PLAN	MONTHLY COST
■80–120/day	6 eggs (■35) + 100g soy chunks dry (■20) + 500ml milk (■30) + 200g dal from mess	Rice/chapati from mess	Groundnut oil (cooking), peanuts 50g	■2,400–3,600
■120–200/day	Above + 100g chicken 3x/week (■25–30) OR canned tuna (■50) 2x/week	Above + 100g oats (■15)	Peanut butter 30g/day	■3,600–6,000
■200–350/day	Full protein plan: eggs + chicken + milk + soy + whey (half scoop)	Oats + rice + banana	Peanut butter + ghee	■6,000–10,500

Hostel Training Plan (No Gym)

DAY	WORKOUT	VOLUME	NOTES
Monday	Push: Push-ups (3×max), Pike push-ups (3×10), Dips on chair (3×max), Diamond push-ups (3×10)	3–4 exercises	Progressive: add reps/harder variants weekly
Tuesday	Pull: Table rows (3×15), Doorframe rows (3×12), Band pull-aparts (3×20), Dead hangs (3×30s)	3–4 exercises	If no equipment: focus on holds and static pulls
Wednesday	Legs: Squats (4×20), Bulgarian split squats (3×12), Glute bridges (3×20), Jump squats (3×15)	4 exercises	Progressively load via pause squats, single-leg
Thursday	Rest / Active recovery	Walk 30 min	Foam roll, stretch
Friday	Full Body Circuit: Squat + Push-up + Row + Plank × 5 rounds	5 circuits	Increase rounds weekly
Saturday	Arms/Core: Chin-ups (3×max), Concentration curl (band), Planks, Hollow holds, Leg raises	3–4 exercises	
Sunday	Rest	Active recovery	Meal prep for week

Time-Efficient Nutrition Strategy

CHALLENGE	SOLUTION
CHALLENGE	SOLUTION
No time to cook	Batch cook Sunday: 2kg chicken + 2kg rice + vegetables. 7-day meal base done in 1 hr.
Skippping breakfast	Overnight oats (prepare in 5 min at night): 100g oats + 300ml milk + 1 banana + 30g whey protein
No time for post-workout meal	Pre-pack: 1 whey scoop in shaker + banana in bag. 30 sec prep, eat at gym
Office lunch (canteen)	Priority order: protein (chicken/paneer/eggs) → vegetables → carbs. Always take protein first.
Evening hunger after work	Pre-made smoothie bag in freezer: banana + milk + peanut butter + protein powder. Blend in 2 min.
Weekend only cooking	Cook 10 kg rice (store in containers) + 3 kg chicken + masala. Reheat daily.

Busy Professional Training (4 days/week, 45–60 min sessions)

SESSION	WORKOUT	TIME
Mon – Upper Strength	Bench 4×5, Row 4×5, OHP 3×8, Pull-ups 3×max	45 min
Tue – Lower Strength	Squat 4×5, RDL 3×8, Leg press 3×12, Calf raise 3×20	45 min
Thu – Upper Hypertrophy	DB Press 4×10, Lat pulldown 4×12, Lateral raise 3×15, Curl 3×12, Tri ext 3×12	50 min
Sat – Lower Hypertrophy	Leg press 4×12, Leg curl 3×12, Lunge 3×10, Hip thrust 3×15, Ab circuit	50 min

[R101] Skinny beginners: aggressive calorie surplus (+400–500 kcal). Liquid calories are their friend.

[R102] Hardgainers: ALWAYS audit actual food intake first. 90% of hardgainers are under-eating, not under-responding.

[R103] Female athletes: same training principles. Adjust for menstrual cycle. Program heavy weeks in follicular phase.

[R104] Female athletes: minimum 15% BF. Iron supplementation if vegetarian or low meat intake.

[R105] Vegetarian athletes: combine complementary proteins. Soy chunks are the single best budget protein.

[R106] Hostel students: batch-cooked soy chunks + mess dal + eggs + milk = complete protein foundation.

[R107] Busy professionals: batch cooking on weekends + liquid shakes = compliance solution.

[R108] No gym = no problem for beginners. Bodyweight progressive overload is fully adequate for 6–12 months.

[R109] Special populations always need lifestyle-first approach. Supplement after lifestyle is optimized.

[R110] Never generate a one-size-fits-all plan. Always consider population-specific constraints.

MODULE 12

COMMON MUSCLE GAIN MISTAKES

Errors, diagnostic logic, correction strategies, and troubleshooting

MISTAKE 1: INSUFFICIENT PROTEIN INTAKE

CATEGORY	DETAIL
WARNING SIGNS	Not hitting 1.6g/kg minimum Eating 1–2 large meals per day Protein concentrated in one meal
Consequence	Suboptimal MPS, muscle loss risk, poor recovery
Correction	Audit protein intake for 3 days Target 30–50g protein per meal Distribute across 4–5 meals Add soy chunks / eggs / whey to fill gaps
AI Rule	IF protein <1.6 g/kg → INCREASE before any other adjustment

MISTAKE 2: INSUFFICIENT CALORIE INTAKE (Most Common)

CATEGORY	DETAIL
WARNING SIGNS	Not gaining weight despite 'eating a lot' Scale weight stagnant for 3+ weeks Constant fatigue during training
Consequence	Zero muscle gain. Body uses protein for energy instead of muscle building.
Correction	Track every meal using app for 7 days Add 300–500 kcal to current intake Use liquid calories (milk, smoothies) Eat 5–6 meals per day
AI Rule	IF scale not moving for 2 weeks on bulk → ADD 200 kcal immediately

MISTAKE 3: PROGRAM HOPPING

CATEGORY	DETAIL
WARNING SIGNS	Changed programs 3+ times in 3 months Bored with current plan after 3–4 weeks Always starting 'new program' found online
Consequence	No adaptation, no progressive overload continuity, no results
Correction	Commit to ONE program for minimum 12 weeks Track lifts. Results are in numbers, not feelings. Understand that adaptation takes time and boring consistency
AI Rule	IF athlete wants to change program <12 weeks in → Review their progress logs first

MISTAKE 4: EXCESSIVE CARDIO DURING BULK

CATEGORY	DETAIL
WARNING SIGNS	Running/cycling 5–7 days/week during muscle gain phase Calorie surplus being eaten by cardio volume Not gaining weight despite eating surplus
Consequence	Cardio burns surplus calories needed for muscle gain. Competes with recovery.
Correction	Reduce cardio to 1–2 sessions/week, 20–30 min moderate intensity IF fat gain occurs: add cardio back gradually Consider step count goal instead of cardio sessions
AI Rule	IF doing >3 cardio sessions/week during bulk AND not gaining → eliminate cardio first

MISTAKE 5: POOR SLEEP

CATEGORY	DETAIL
WARNING SIGNS	Sleeping <6 hrs per night Irregular sleep schedule Alcohol consumption affecting sleep quality
Consequence	GH suppression, testosterone reduction, impaired MPS, poor recovery, muscle loss

CATEGORY	DETAIL
Correction	Set non-negotiable sleep time No screens 60 min before bed No caffeine after 2 PM Magnesium glycinate 300mg pre-sleep
AI Rule	IF sleeping <7 hrs → Address sleep BEFORE adding training volume or supplements

MISTAKE 6: INCONSISTENT TRAINING

CATEGORY	DETAIL
WARNING SIGNS	Missing >2 sessions/week Training 'when motivated' Taking 2+ weeks off without reason
Consequence	No progressive overload, detraining within 2–3 weeks, wasted nutrition
Correction	Set minimum 3 training days/week as non-negotiable Track attendance over 4 weeks Shorter 30-min sessions are better than zero sessions
AI Rule	IF missing >2 sessions/week → Fix schedule adherence before optimizing nutrition

MISTAKE 7: NO PROGRESSIVE OVERLOAD

CATEGORY	DETAIL
WARNING SIGNS	Same weight, same reps for months No training log or records Relies on 'feeling' not numbers
Consequence	No adaptation signal → No muscle growth
Correction	Implement training log immediately Target +2.5 kg or +1 rep every session (beginner) Apply double progression: reps then load
AI Rule	IF no progressive overload in 4 weeks → Entire plan is ineffective regardless of nutrition

MISTAKE 8: UNREALISTIC EXPECTATIONS

CATEGORY	DETAIL
WARNING SIGNS	Expects 5 kg muscle in 1 month Expects visible muscle by week 2 Compares to steroid-enhanced social media physiques
Consequence	Dropout. Program hopping. Supplement overspending.
Correction	Set realistic expectations in onboarding (Module 1 timeline) Explain natural vs enhanced athlete differences Track strength gains as short-term marker
AI Rule	IF athlete discouraged at 4 weeks → Show strength progress data. Muscle follows strength.

MISTAKE 9: NEGLECTING COMPOUND MOVEMENTS

CATEGORY	DETAIL
WARNING SIGNS	Program is 80% isolation work (curls, extensions, flies) No squat, deadlift, or barbell press Training 'arms only' or mirror muscles
Consequence	Suboptimal hormonal response, poor muscle mass distribution, inefficient time use
Correction	Rebuild program with compound-first approach Squat + Deadlift + Bench + Row + OHP as foundation Add isolation only after compound work
AI Rule	IF program has <50% compound movements → Restructure immediately

MISTAKE 10: FEAR OF DIETARY FAT

CATEGORY	DETAIL
WARNING SIGNS	Eating <20% calories from fat Avoiding eggs, ghee, nuts, dairy fat Testosterone issues, hormonal imbalances
Consequence	Testosterone suppression, poor hormone function, impaired vitamin absorption
Correction	Increase fat to 25–30% of total calories Include ghee, eggs, nuts, olive oil, whole milk Reassure athlete that dietary fat ≠ body fat
AI Rule	IF dietary fat <20% of calories → Immediate increase to minimum 20%

MISTAKE 11: SKIPPING POST-WORKOUT NUTRITION

CATEGORY	DETAIL
WARNING SIGNS	Training then going home, eating 2+ hrs later No post-workout protein Using 'I'll eat dinner later' as plan
Consequence	Missed peak MPS window, slower glycogen recovery
Correction	Pre-pack post-workout shake in gym bag Even banana + milk = adequate quick fix Protein within 60 min is non-negotiable during bulk
AI Rule	IF athlete routinely skips post-workout nutrition → Set reminder system + pre-packed option

MISTAKE 12: DIRTY BULKING (Unnecessary Fat Gain)

CATEGORY	DETAIL
WARNING SIGNS	Eating anything and everything Gaining >1 kg/week consistently BF rising rapidly above optimal range
Consequence	Fat gain requiring future cut, insulin resistance risk, poor training performance
Correction	Track calories. Reduce surplus to +200–350 kcal Higher protein (2g/kg) to maximize muscle retention ratio Cut back to lean bulk strategy
AI Rule	IF gaining >0.6 kg/week → Reduce calories by 200 kcal immediately

IF NOT GAINING WEIGHT for 2 weeks → THEN Step 1: Increase calories +200 kcal. Step 2: Track food log 7 days. Step 3: Reduce cardio.

IF GAINING FAT too fast (>0.6 kg/week) → THEN Reduce calories -200 kcal. Increase protein to 2.2 g/kg. Maintain training volume.

IF STRENGTH DECLINING on bulk → THEN Check: sleep, calories (below TDEE?), stress, deload needed?

IF POOR RECOVERY (always sore) → THEN Increase sleep to 8+ hrs. Reduce training volume by 20%. Add post-workout nutrition.

IF PLATEAU (no strength gain for 4 weeks) → THEN Deload 1 week. Return with periodized program. Increase volume by 2 sets.

IF LOW ENERGY in training → THEN Check pre-workout carbs. Check sleep. Check iron/vitamin D (females). Add caffeine if needed.

IF GI DISTRESS from protein shakes → THEN Switch whey concentrate → whey isolate. Reduce dose. Space protein meals further.

IF NOT HITTING PROTEIN TARGET (vegetarian) → THEN Add soy chunks 100g dry/day. 52g protein at █20. Non-negotiable base.

IF LOSING MUSCLE on bulk (measurement) → THEN Protein below 1.6 g/kg? Sleep below 7 hrs? Training inconsistency? Check all three.

IF MOTIVATION LOW → THEN Re-assess goals. Show progress data. Switch program (within same principles). Accountability partner.

[R111] Always diagnose mistake before prescribing solution. Symptoms can have multiple causes.

[R112] Insufficient calories is the #1 reason for no muscle gain. Always check this first.

[R113] Program hopping is a behavior problem, not a program problem. Address mindset.

[R114] Progressive overload is measurable. No progress data = no coaching possible.

[R115] Sleep issues override all other coaching advice. Sleep first, then optimize everything else.

[R116] Excessive cardio during bulk is a silent muscle gain killer. Quantify cardio volume.

[R117] Fat phobia leads to hormonal dysfunction. Educate and correct dietary fat intake.

[R118] Post-workout nutrition missed >3x/week is a system problem. Create a simple system.

[R119] Dirty bulk: maximum 4 weeks acceptable before reassessing body composition.

[R120] If all variables check out (calories, protein, training, sleep) and still no gain: reassess TDEE. May need more calories.

MODULE 13

COACHING RULES MASTER SECTION

300+ coaching rules, decision logic, plateau handling, and progress tracking

NUTRITION RULES

- [R121] Total daily protein is the most important nutrition variable for muscle gain.
- [R122] Caloric surplus is required for muscle gain. No surplus = no muscle gain (except beginners/recomp).
- [R123] Protein distribution: spread across 4–5 meals for maximum MPS stimulation.
- [R124] Pre-workout: carbs + moderate protein, low fat, 1.5–2 hrs before training.
- [R125] Post-workout: 30–50g fast protein + 60–120g fast carbs within 60 min.
- [R126] Creatine monohydrate: 3–5g daily. No loading needed. Take consistently.
- [R127] Hydration: 35–40ml/kg BW per day. Add 500–750ml per training hour.
- [R128] Dietary fat minimum: 20% of total calories. Never go below for testosterone protection.
- [R129] Omega-3 target: 2–4g EPA+DHA per day from food + supplements.
- [R130] Fiber target: 25–35g/day from vegetables, legumes, whole grains.
- [R131] Micronutrient priority: Vitamin D, Iron (females), Zinc, Magnesium, B12 (vegans).
- [R132] Do not restrict any macronutrient during muscle gain phase.
- [R133] Whole food > Supplements. Food matrix provides nutrients supplements cannot replicate.
- [R134] Pre-sleep casein protein (30–40g): improve overnight MPS for bulking athletes.
- [R135] Calorie tracking: mandatory for first 4–6 weeks until athlete learns portion sizes.

TRAINING RULES

- [R136] Every training session must implement some form of progressive overload.
- [R137] Track all lifts in every session. No data = no coaching.
- [R138] Compound movements first in every session. Isolation movements after.
- [R139] Beginner: 3 full body sessions/week. Intermediate: 4–5 sessions. Advanced: 5–6.
- [R140] Volume: 10–20 sets per muscle group per week. Increase by 2 sets per 4-week block.
- [R141] Rep range: 6–12 primary. Include 4–6 heavy sets and 12–20 pump sets for completeness.
- [R142] RIR: Work within RIR 1–3 for all hypertrophy work. Near-failure mandatory.
- [R143] Rest 2–3 min for heavy compound work. 60–90 sec for isolation/pump work.
- [R144] Deload every 4–6 weeks. Reduce volume by 50%, maintain intensity.
- [R145] Minimum training frequency: 2x/week per muscle group for intermediate-advanced.
- [R146] Form check: video record major lifts monthly. Form breakdown = injury risk.
- [R147] Unilateral work: include single-leg and single-arm exercises to address imbalances.
- [R148] Warm-up: 5–10 min general + 2–3 warm-up sets for first compound exercise.
- [R149] Cool down: 5–10 min static stretching for trained muscle groups.
- [R150] Never skip legs. Lower body provides systemic anabolic hormonal environment.

RECOVERY RULES

- [R151] Sleep is the single most important recovery factor. Minimum 7 hrs. Optimal 8–9 hrs.
- [R152] Consecutive days of heavy compound work without adequate sleep = injury and regression risk.
- [R153] Cortisol management: training load must match stress tolerance, not just physical capacity.
- [R154] Rest days are growth days. Muscle grows during recovery, not during training.
- [R155] Active recovery: light walking, yoga, swimming improves blood flow without taxing CNS.
- [R156] Massage and foam rolling: reduce soreness perception and improve ROM.

- [R157] Alcohol is incompatible with optimal muscle gain. Zero tolerance during bulk.
- [R158] If DOMS lasts >72 hrs, volume was too high or recovery was insufficient.
- [R159] OTS: full rest from training, not deload. Differentiate overtraining from overreaching.
- [R160] CNS fatigue: plan heavy sessions on days with adequate prior sleep and nutrition.

ASSESSMENT RULES

- [R161] Initial assessment: bodyweight, height, BF estimate, training experience, diet type, budget, schedule.
- [R162] Weekly check-in: scale weight (3-day average), training performance, energy, sleep quality.
- [R163] Monthly assessment: body measurements (chest, waist, arms, legs), progress photos, strength benchmarks.
- [R164] 3-day average scale weight prevents misleading daily fluctuations from guiding decisions.
- [R165] Body weight trending up 0.1–0.5 kg/week = successful lean bulk.
- [R166] Strength benchmarks for 6-month progress: Squat +20%, Bench +20%, Deadlift +25%.
- [R167] BF estimation: use calipers, DEXA (gold standard), or progress photos. Avoid cheap bioimpedance.
- [R168] Training log review: at least weekly. Look for strength plateau, volume response, recovery trends.
- [R169] Food log: mandatory for first month. Optional after athlete demonstrates awareness.
- [R170] If athlete is not compliant with tracking: simplify to hand-portion tracking. Adherence > Precision.

A. Initial Athlete Assessment Decision Tree

- IF athlete BMI <18.5 AND training <1 year → THEN Aggressive lean bulk (+400 kcal), 3-day full body, 1.8g/kg protein*
- IF athlete BMI 18.5–24.9 AND training <1 year → THEN Moderate lean bulk (+250–300 kcal), 3-day full body, 1.8g/kg protein*
- IF athlete BMI 25–29.9 AND training <1 year → THEN Recomp (TDEE ±0), 3–4 day full body, 2.0g/kg protein, slight deficit optional*
- IF athlete BMI >30 AND training <1 year → THEN Mild deficit (TDEE –300 kcal), 3-day full body, 2.2g/kg protein, high protein satiety*
- IF athlete is INTERMEDIATE AND BF <15% (M) / <23% (F) → THEN Lean bulk (+200–300 kcal), Upper-Lower or PPL split, 2.0g/kg protein*
- IF athlete is INTERMEDIATE AND BF >20% (M) / >28% (F) → THEN Mini-cut first (8 weeks, TDEE –300 kcal), then transition to lean bulk*
- IF athlete is ADVANCED AND <10% BF (M) → THEN Lean bulk (+150–200 kcal). Slow and controlled. 2.0–2.2g/kg protein.*
- IF athlete is ADVANCED AND >15% BF (M) → THEN Recomp or mini-cut. Advanced athletes gain minimal muscle in surplus when BF is high.*

B. Weekly Adjustment Decision Tree

- IF weight UP 0.1–0.3 kg/week → THEN Stay the course. Lean bulk on track.*
- IF weight UP 0.3–0.6 kg/week AND strength increasing → THEN Acceptable. Monitor BF. Reduce if waist gaining >1 cm/month.*
- IF weight UP >0.6 kg/week → THEN Excess fat gain. REDUCE calories 200 kcal.*
- IF weight UNCHANGED for 2 weeks → THEN INCREASE calories 150–200 kcal.*
- IF weight DOWN despite surplus on tracking → THEN Reassess tracking accuracy. Recalculate TDEE. Stress/sleep check.*
- IF strength INCREASING + weight stable → THEN Recomp occurring. Good for BF >15% athletes. Continue.*
- IF strength DECLINING on bulk → THEN Check: sleep quality, calorie intake accuracy, deload need, stress.*
- IF recovery POOR (3+ days DOMS, fatigue) → THEN Reduce training volume 20%, increase calories 100 kcal, prioritize sleep.*
- IF athlete MISSES 2+ sessions this week → THEN Investigate reason. System fix. Do not add volume next week.*
- IF athlete REPORTS low energy, mood, libido → THEN Check: sleep, calories, overall stress, deload need, blood work.*

C. Plateau Handling Decision Tree

- IF STRENGTH PLATEAU for 2 weeks (beginner) → THEN Check: sleep, nutrition, form. If okay: try 1 deload week then reload heavier.*
- IF STRENGTH PLATEAU for 4 weeks (beginner) → THEN Program change needed. Switch from LP to wave periodization. Add variety.*
- IF STRENGTH PLATEAU (intermediate) → THEN Implement periodization: accumulation → intensification → deload. 3-4 week blocks.*
- IF WEIGHT PLATEAU on bulk for 3 weeks → THEN Increase calories +200. Reassess tracking. Add 1 meal. Increase pre-workout carbs.*

IF MUSCLE GAIN PLATEAU despite progress in training → THEN Assess: sleep 8+, protein 2g/kg, surplus adequate? Add volume 2 sets.

IF ALL VARIABLES CORRECT and still no progress → THEN Consider cyclic approach: 4 weeks bulk + 1 week maintenance. Gives metabolic reset.

IF PERSISTENT PLATEAU >3 months → THEN Full program overhaul. New split, new exercises, new periodization. Body has adapted to current stimulus.

Weekly Check-In Protocol

METRIC	HOW TO MEASURE	FREQUENCY	GOOD SIGNAL	BAD SIGNAL
Scale Weight	Morning, post-toilet, fasted. Average 3 consecutive days.	Daily (average weekly)	+0.1–0.5 kg/week	Losing weight on bulk OR gaining >0.6 kg/week
Training Log	Record: exercise, sets, reps, weight each session	Every session	Strength increasing or volume increasing	Same weights/reps for 3+ weeks
Energy in Training	Rate 1–10 during session	Every session	7–9/10 consistent	Below 6/10 3+ sessions in a row
Sleep Quality	Hours + wake-up feeling (1–10)	Daily	7+ hrs, feeling rested	<7 hrs, still tired on waking
Recovery	Rate soreness and fatigue 1–10	Daily	3–4/10 mild soreness	7+/10 chronic, lingering soreness
Hunger	Rate 1–10	Daily	Moderate (5–7), manageable	Always extremely hungry (under-eating) or never hungry (low activity)
Body Measurements	Chest, waist, hips, arms, legs in cm	Monthly	Arms/chest increasing, waist stable	Waist increasing rapidly (+2 cm/month)
Progress Photos	Same lighting, same pose, same time of day	Monthly	Visible fullness, muscle definition	No visual change over 3 months

Linear Periodization (Beginners – First 12 months)

- Add weight to bar every session (or every 2 sessions for upper body)
- If miss rep target → stay at same weight next session
- If 3 consecutive fails → deload 10% and rebuild
- Program: 3×5 for compounds (beginner strength phase)

Double Progression (Intermediate)

- Set target rep range (e.g., 3×8–12)
- Achieve top of range (3×12) → increase weight next session
- Start at bottom of rep range with new weight (e.g., 3×8)
- Climb range again. Takes 2–4 weeks per increment.

Block Periodization (Intermediate-Advanced)

BLOCK	WEEKS	SETS × REPS	LOAD	GOAL
Accumulation	3–4 weeks	4×12–15	60–70% 1RM	Volume, metabolic stress, sarcoplasmic
Intensification	3–4 weeks	4×4–6	80–90% 1RM	Strength, myofibrillar density
Realization	1–2 weeks	3×1–3	90–100% 1RM	Test max, peak performance
Deload	1 week	3×8–10	50–60% 1RM	Recovery, restoration

[R171] Never generate a training plan without knowing the athlete's available equipment.

[R172] Never generate a nutrition plan without knowing the athlete's food budget.

- [R173] If athlete has a food allergy, substitute before generating any meal plan.
- [R174] Always confirm if athlete is vegetarian, vegan, or non-veg FIRST before meal planning.
- [R175] Female athletes approaching menstruation (days 21–28): reduce training volume by 15–20%.
- [R176] Ectomorphs: calorie surplus is king. Training comes second. Eat more. Always.
- [R177] Endomorphs: lean bulk only. Never dirty bulk. Monitor fat gain weekly.
- [R178] If athlete is student with exam season: reduce training volume 30%. Sleep priority.
- [R179] Do not prescribe >6 training days/week to any natural athlete.
- [R180] Minimum 1 full rest day per week from all structured exercise.
- [R181] Pre-workout supplement with caffeine: limit to 5 days/week maximum.
- [R182] Creatine: do not require rest days. Daily continuous use is correct.
- [R183] BCAA supplements: do not recommend if total protein target is met. Waste of money.
- [R184] Pre-workout meal timing: 1.5–2 hrs for full meal. 30–45 min for light snack.
- [R185] Post-workout window: 60 min is ideal. 2 hrs is acceptable. Beyond 2 hrs: diminishing returns.
- [R186] Intra-workout nutrition: only needed for sessions >90 min.
- [R187] Endurance athletes who also do resistance training: additional 0.5 g/kg protein required.
- [R188] Combine protein sources at each meal for improved amino acid profile completeness.
- [R189] Cold weather: TDEE increases slightly (+5–10%). Account in calorie target.
- [R190] Hot weather (summer in India): electrolyte needs increase. Add sodium to post-workout.
- [R191] Ramadan fasting athletes: shift training window to post-iftar. Meal timing adapts to fast.
- [R192] If athlete reports joint pain: add fish oil, reduce impact training, assess form.
- [R193] For knee pain: reduce squat depth temporarily. Switch to leg press. Do not stop training.
- [R194] For lower back pain: reduce deadlift load 20%. Assess hip hinge form. Add core work.
- [R195] For shoulder pain: avoid overhead pressing. Switch to neutral grip, reduce range of motion.
- [R196] Injury = maintain nutrition at maintenance. Do not deficit during injury. Maintain protein.
- [R197] Athlete returning after injury: reduce training volume 30%. Rebuild over 4 weeks.
- [R198] Intermittent fasting (16:8): acceptable if 4+ protein meals fit in 8hr eating window.
- [R199] Early morning training (before 7 AM): protein shake pre-training minimum. Full meal after.
- [R200] Meal prep containers, pre-cooked protein, and batch cooking are not optional for success. They are the system.
- [R201] If athlete has poor digestion: recommend digestive enzymes with high-protein meals.
- [R202] Probiotics may improve nutrient absorption and protein digestion. Recommend if GI issues present.
- [R203] Athletes eating very high protein (>2.5 g/kg): ensure fluid intake 40ml/kg minimum.
- [R204] Protein powder is not absorbed differently than food protein. Quality is what matters.
- [R205] Warm up sets do NOT count toward working set volume. Start tracking after warm-up.
- [R206] Volume cycling: higher volume in hypertrophy phases, lower volume in strength phases.
- [R207] Supersets and giant sets: reduce rest time, increase density, add variety. Good for time-crunched athletes.
- [R208] Unilateral exercises (single-leg, single-arm): include 1–2 per session for balance.
- [R209] Core training: functional core stability (planks, dead bugs) > crunches for athletes.
- [R210] Calves: high frequency response. Train 3–4x/week. High rep ranges (15–20+).
- [R211] Forearms develop primarily through grip work during compound pulling. Specific work usually unnecessary.
- [R212] Neck training: include neck work for combat athletes and contact sport athletes.
- [R213] If athlete has plateaued, switching to a higher rep range temporarily can reignite growth.
- [R214] Drop sets: effective for isolation exercises. Use as finisher, not primary loading.
- [R215] Rest-pause sets: advanced technique. Not for beginners. High CNS demand.
- [R216] Blood flow restriction (BFR) training: effective for injury recovery and low-load hypertrophy.
- [R217] Always assess: is athlete consistent? 70% adherent to a perfect plan beats 100% adherent to a poor plan.
- [R218] Alcohol + training day: alcohol before bed blunts overnight MPS by 24%.
- [R219] Social situations: plan protein intake first at any social meal. Protein + vegetables first.
- [R220] Restaurant eating: order grilled/tandoori protein. Dal makhani + roti is good base.
- [R221] Travel: protein bars + nut packets + protein powder sachets are travel nutrition kit.
- [R222] Lean bulk rate: 0.25–0.5 kg/month expected. Do not panic in first 2 weeks (water weight adjustment).
- [R223] Muscle weight gain vs water weight: first 1–2 weeks of creatine use will show 1–2 kg apparent gain (water in muscle). This is normal.
- [R224] Waist measurement: primary fat gain indicator. If waist gaining >1 cm/month, reduce surplus.
- [R225] Progress photos: front, side, back relaxed + flexed. Same time, same day each month.

- [R226] Confidence in the process is a coaching skill. Athletes need reassurance backed by data.
- [R227] Long-term perspective: 1 year of consistent lean bulking = 6–10 kg muscle. 5 years = transformative physique.
- [R228] Motivation fluctuates. Systems and habits do not. Build systems, not motivation.
- [R229] NEAT (Non-Exercise Activity Thermogenesis): tracking steps is an excellent proxy. Target 8,000–10,000 steps/day.
- [R230] High NEAT athletes (step count >12,000/day) need higher TDEE estimates. Use 1.7–1.9 multiplier.
- [R231] Sedentary athletes with desk jobs: extra NEAT activities (standing desk, walking meetings) can add 200–400 kcal burn.
- [R232] Glycemic index only matters around training. Rest of day: total carbs and fiber quality matter more.
- [R233] Time-restricted eating ≠ intermittent fasting. Muscle gain is possible with either if protein targets met.
- [R234] Food timing optimization is a tier-3 priority. Total daily intake is always tier-1.
- [R235] Meal replacement shakes: acceptable occasionally. Not as primary nutrition strategy.
- [R236] Vegetarian protein ranking (cost-effectiveness): Soy chunks > Eggs > Paneer > Milk > Greek yogurt > Whey.
- [R237] Non-vegetarian protein ranking (cost-effectiveness): Eggs > Tuna (canned) > Chicken > Fish > Beef.
- [R238] When optimizing for a vegetarian athlete, soy chunks (nutri-nuggets) should be the anchor protein source.
- [R239] Gluten intolerance: replace wheat with rice, oats (certified GF), potato, quinoa. Protein sources unaffected.
- [R240] Nut allergy: replace nut-based fat sources with seeds (sunflower, pumpkin, chia), olive oil, ghee.
- [R241] Egg allergy: replace with soy chunks, paneer, tofu, legumes, dairy (if tolerated), whey (if no egg component).
- [R242] Dairy allergy: use coconut milk, soy milk, oat milk (lower protein). Rely on soy protein isolate.
- [R243] Supplement label reading: per 100g protein content >70% = quality protein. Proprietary blends = avoid.
- [R244] Creatine: store in dry location. Mix with water or juice. Do not mix with hot liquids.
- [R245] Mass gainers: check label. Most are expensive maltodextrin + low quality protein. DIY is better.
- [R246] DIY mass gainer: 100g oats + 50g peanut butter + 2 scoops whey + 500ml whole milk = ~900 kcal, 70g protein.
- [R247] Whey protein brand selection: check amino acid profile (spiking with amino acids like taurine/glycine is a red flag).
- [R248] Indian gym environment: most people overtrain legs or neglect them entirely. Prescribe squat as foundation.
- [R249] Social media comparison: Instagram/YouTube physiques 95% enhanced. Educate athletes on natural timelines.
- [R250] Three-month check: if no visual progress at 3 months, check photos side-by-side vs beginning. Progress is often invisible day-to-day.

OBSERVATION	CALORIES	PROTEIN	TRAINING	SLEEP/RECOVERY
On track (0.1–0.4 kg gain/week, strength up)	No change	No change	No change	No change
Not gaining weight 2+ weeks	+150–200 kcal	No change	Check volume adequate	Check stress
Gaining too fast (>0.6 kg/week)	–150–200 kcal	Increase 0.2 g/kg	No change	No change
Strength declining despite weight gain	No change	+0.2 g/kg	Add deload	Prioritize sleep
Losing weight on surplus	Audit tracking + +200 kcal	+0.2 g/kg	Reduce cardio	Check sleep
Poor recovery, excessive fatigue	+100 kcal	+0.2 g/kg	Reduce volume 20%	Sleep +1 hr target
Low training energy	+50–100g carbs pre-workout	No change	Reduce intensity temporarily	Check iron/vitamin D
GI distress	No change	Spread doses	No change	No change
Motivation low, mood poor	No change	No change	Program variety, deload	Sleep + stress audit
Traveling / disrupted schedule	Maintain protein minimum	Protein priority	Minimal effective training	Hotel room workout

Male Athletes – Relative to Bodyweight

LIFT	BEGINNER (3mo)	INTERMEDIATE (1yr)	ADVANCED (3yr)	ELITE (5yr+)
Squat	0.5× BW	1.0× BW	1.5× BW	2.0× BW

LIFT	BEGINNER (3mo)	INTERMEDIATE (1yr)	ADVANCED (3yr)	ELITE (5yr+)
Deadlift	0.75× BW	1.25× BW	1.75× BW	2.5× BW
Bench Press	0.5× BW	0.75× BW	1.0× BW	1.25× BW
Overhead Press	0.35× BW	0.5× BW	0.65× BW	0.8× BW
Barbell Row	0.4× BW	0.75× BW	1.0× BW	1.25× BW
Pull-ups	1 rep	10 reps BW	20 reps BW	+25kg weighted

Female Athletes – Relative to Bodyweight

LIFT	BEGINNER (3mo)	INTERMEDIATE (1yr)	ADVANCED (3yr)	ELITE (5yr+)
Squat	0.35× BW	0.75× BW	1.0× BW	1.5× BW
Deadlift	0.5× BW	1.0× BW	1.25× BW	1.75× BW
Bench Press	0.25× BW	0.5× BW	0.75× BW	1.0× BW
Overhead Press	0.2× BW	0.35× BW	0.5× BW	0.65× BW
Barbell Row	0.25× BW	0.5× BW	0.75× BW	1.0× BW
Pull-ups	0 reps	1–5 reps	8–12 reps	15+ reps

A. Daily Macro Template at Common Calorie Targets

CALORIE TARGET	PROTEIN (g) [2g/kg @ 75kg]	CARBS (g) [50% cal]	FAT (g) [25% cal]	EXAMPLE ATHLETE
2,000 kcal	150g (no less)	250g	56g	Female, moderate activity
2,500 kcal	150g	313g	69g	Male beginner, light activity
3,000 kcal	160–180g	375g	83g	Male intermediate, mod activity
3,500 kcal	170–200g	438g	97g	Male active, lean bulk
4,000 kcal	180–220g	500g	111g	Very active/large male, hardgainer

B. Sample Meal Plan Template (Non-Vegetarian, 3000 kcal, 180g Protein)

MEAL	FOOD	CALS	PROTEIN	CARBS	FAT
Meal 1 (8AM)	4 whole eggs + 2 egg whites + 100g oats + 300ml milk	650	45g	72g	22g
Meal 2 (12PM)	200g chicken breast + 250g rice (cooked) + 200g dal + sabzi	750	55g	95g	14g
Meal 3 (3PM) Snack	100g roasted chana + 1 banana + 300ml milk	450	25g	62g	8g
Pre/Post Training (5–7PM)	1.5 scoop whey + 300ml milk + banana	450	48g	52g	8g
Meal 4 (8PM)	200g fish/paneer + 200g rice + vegetables + 1 tsp ghee	700	45g	82g	18g
Meal 5 (10PM)	200g paneer + 100ml milk (casein)	370	30g	10g	22g
TOTAL	—	3,370	248g	373g	92g

C. Sample Meal Plan Template (Vegetarian, 3000 kcal, 160g Protein)

MEAL	FOOD	CALS	PROTEIN	CARBS	FAT
Meal 1 (8AM)	4 whole eggs + 100g oats + 300ml whole milk	620	44g	65g	22g
Meal 2 (12PM)	150g paneer (cooked) + 250g rice + 200g dal + vegetables	750	42g	96g	18g
Meal 3 (3PM)	100g soy chunks cooked (~52g protein dry) + 2 roti	500	38g	58g	6g
Post Training (6PM)	1.5 scoop whey + 300ml milk + banana	450	46g	52g	8g

MEAL	FOOD	CALS	PROTEIN	CARBS	FAT
Meal 4 (8PM)	200g paneer + 200g rice + sabzi + 1 tsp ghee	700	32g	76g	25g
Meal 5 (10PM)	300g Greek yogurt + 30g peanut butter	420	26g	20g	24g
TOTAL	—	3,440	228g	367g	103g

D. Training Plan Template (Intermediate, Upper-Lower, 4 Days)

SESSION	EXERCISE	SETS × REPS	NOTES
Upper A (Mon)	Bench Press 4×6–8 Barbell Row 4×6–8 OHP 3×8–10 Lat Pulldown 3×10–12 Curl 3×12 Tri Ext 3×12	~24 sets	Strength focus. 2–3 min rest compounds.
Lower A (Tue)	Squat 4×5–8 Romanian DL 3×8–10 Leg Press 3×12 Leg Curl 3×12 Calf Raise 4×15–20	~17 sets	Strength focus. Squat heavy.
Rest (Wed)	Active recovery: walk 30 min + stretch	—	Do NOT train
Upper B (Thu)	Incline DB Press 4×10–12 Cable Row 4×12 Lateral Raise 4×15 Face Pull 3×15 Hammer Curl 3×12 Skull Crusher 3×12	~24 sets	Volume/hypertrophy focus. 60–90s rest.
Lower B (Fri)	Deadlift 4×4–6 Bulgarian Split Squat 3×10 Hip Thrust 4×12 Leg Ext 3×15 Leg Curl 3×12 Calf Raise 3×15	~20 sets	DL heavy, accessories volume.
Rest (Sat + Sun)	Full rest or 1 day active recovery	—	Recovery priority

- [R251] The hierarchy of muscle gain: Calories > Protein > Training > Sleep > Supplements > Everything else.
- [R252] If any of the Big 4 (Calories, Protein, Training, Sleep) is deficient, results will be compromised.
- [R253] Personalization is the difference between a good plan and a great plan. Always assess before prescribing.
- [R254] Compliance to a mediocre plan > Non-compliance to a perfect plan. Make plans the athlete will follow.
- [R255] One year of consistent training with proper nutrition = transformation most people don't achieve in 5 years.
- [R256] AI Coach role: assessment → plan generation → adjustment → troubleshooting → education.
- [R257] Every coaching decision should be backed by at least one of the rules in this knowledge base.
- [R258] RAG system note: when generating advice, always cite the relevant module and rule number for transparency.
- [R259] Athlete safety is first priority. If any health red flag is raised, refer to medical professional.
- [R260] This knowledge base is optimized for Indian athletes (budget, food culture, lifestyle). Adapt as needed for other regions.

END OF ZITLAS AI MUSCLE GAIN BRAIN – VOLUME 1

Total: 13 Modules | 260+ Coaching Rules | 50+ Decision Logic Rules | 40+ Tables
ZITLAS AI Coach System | RAG Knowledge Base | © 2025 ZITLAS